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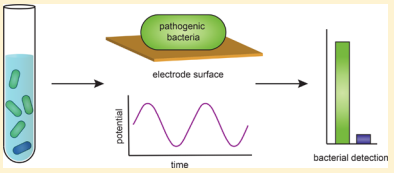
Impedance-Based Detection of Bacteria

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ABSTRACT: Pathogenic bacteria have always posed one of the most serious threats to public health, and continue to be especially dangerous with the rise in antibiotic resistance. The prevalence of these infectious agents necessitates rapid, point-of-care sensors for their detection, identification, and monitoring. Electrochemical sensors are promising for the low-cost monitoring of bacterial growth and the detection of specific microbial species due to the consistency and ease-of-use of impedance measurements. Though the commercialization of these sensors is currently limited, they offer significant promise for detecting pathogens from real-world environments.



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1. INTRODUCTION

Bacteria are ubiquitous both in the environment and in the human body, putting us in contact with countless different species daily. Fortunately, most of these are not harmful to humans, with many even forming a beneficial relationship with their hosts. However, pathogenic strains that lead to human disease do exist and pose a serious threat to public health, especially considering the generally small numbers of bacteria required for infectivity and the ever-increasing rates of antimicrobial resistance in these infections (Figure 1). The most common bacterial-derived diseases are foodborne illnesses, urinary tract infections, sexually transmitted infections, and healthcare-associated infections.

The rapid and sensitive detection of pathogens has long been a goal for point-of-care diagnostic tools and field tests for food- and waterborne pathogens, though the necessary speed and sensitivity required for these sensors to be useful pose a significant challenge. Electrochemical sensors offer all the advantages of laboratory assays (specificity, selectivity, and sensitivity) with the low cost and speed required for point-of-care analyses. Impedance sensors, specifically, have been used for the monitoring and detection of bacteria for over half a century because the measurements are easy to do, often do not require labels, offer compatibility with complex solutions including food and beverages, and yield highly reproducibility results. Herein, we provide an overview of existing electrochemical detection systems for bacteria and then focus on the specifics of the impedance-based detection of pathogenic bacteria.

1.1. Foodborne Illness-Causing Bacteria

Foodborne pathogens are of particular interest to public health, and over 90% of these are bacteria.[1] The Centers for Disease Control and Prevention (CDC) estimates about 9.4 million cases of foodborne illness annually, with these pathogens causing about 50 000 hospitalizations and 1500 deaths.[2] The annual economic cost of the five main bacterial infections alone is nearly $7 billion, as estimated by the U.S. Department of Agriculture.[3] The persistence of these bacteria in food and water supplies around the globe necessitates rapid, inexpensive point-of-care sensors for their detection.

1.2. Urinary Tract Infections

Urinary tract infections (UTIs) are some of the most common causes of bacterial infection worldwide.[4] It is estimated that over 150 million people will contract a UTI each year,[5] leading to 10 million visits to the doctor and 2 million visits to the emergency room annually.[6,7] The economic cost of these infections in the United States tops $3.5 billion per year, with the cost increasing yearly.[4] As the first line of treatment for these infections is antibiotics, many UTIs are now antibiotic resistant, making them more and more difficult to treat. The rapid identification of antibiotic resistances in patient samples would ameliorate improper antibiotic usage.

1.3. Sexually Transmitted Infections

It is estimated that over 1 million new sexually transmitted infections (STIs) occur daily.[8] Bacterial STIs are known to cause serious and long-term health problems, such as the link between syphilis and birth defects,[9] and are also associated with preterm delivery and low baby birth weight, female infertility, and pelvic inflammatory disease.[8] As antimicrobial resistance has increased in the past decade, resistances in STIs have also increased significantly. One of the best-known cases is the emergence of a multidrug-resistant gonorrhea, which will soon become completely untreatable with current antibiotics.[10,11] One key challenge in the fight against the spread of bacterial STIs is the lack of rapid, portable, and inexpensive diagnostics for these diseases. Such devices would enable at-home testing that could prevent the spread of disease.

1.4. Healthcare-Associated Infections

Healthcare-associated infections (HCAIs), or nosocomial infections, are those that occur following medical treatment in a healthcare facility. Nearly 7% of patients in the developed world contract an HCAI,[12] with the incidence of infection in intensive care units (ICUs) now over 50%.[13] According to CDC estimates, the average cost to treat an HCAI is just over $2,000, adding a significant expense to already high healthcare costs. The key types of HCAIs include bloodstream infections, urinary tract infections (UTIs), pneumonia, and surgical site infections.[14,15] Bacterial infections comprise a large fraction of total nosocomial infections,[15,16] and the prevalence of resistance to commonly administered antibiotics again poses a significant challenge.[16] This has led to ineffective initial treatments and the persistence of resistances in healthcare settings. Furthermore, it has been found that the presence of multispecies pathogenic colonies leads to significant biofilm growth[17] that is accompanied by a further increase in antibiotic resistance.[18,19] Both the economic burden and common

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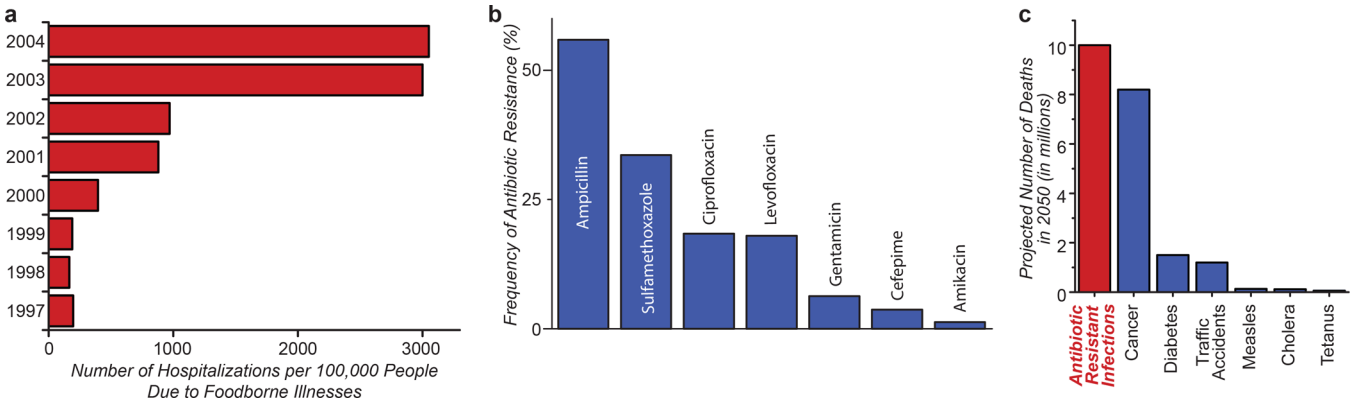


Figure 1. Increasing prevalence of health risks caused by microbial infections. (a) Annual number of hospitalizations (per 100 000 people) due to foodborne illnesses in the United States.[2] (b) Percent of community acquired urinary tract infections with antibiotic resistance in Bahia, Brazil, from 2010 to 2014.[7] (c) Projected number of deaths by cause in 2050. In 2014, 700 000 deaths were due to antimicrobial resistant infections.[9,10]

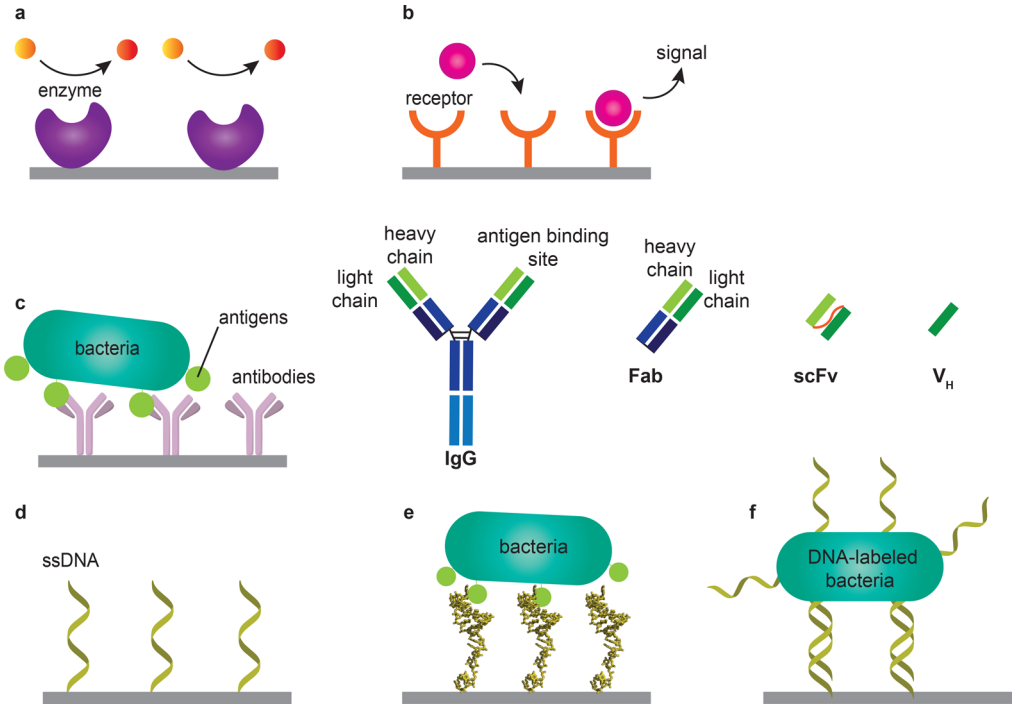


Figure 2. Biorecognition elements. (a) Enzymes are proteins responsible for the chemical conversion of molecules, which can be used as a readout method. (b) Receptors are proteins that bind specific targets, causing a downstream signal. (c) Antibodies are large glycopeptides that have especially high specificity for their targets. Antibody fragments are smaller portions of the full polypeptide that retain their recognition abilities. IgG = immunoglobulin G; Fab = antigen-binding fragment; scFv = single-chain variable fragment; v_H = single-domain antibody fragment. (d) Single-stranded DNA (ssDNA) can either (e) adopt a unique 3D conformation to bind to a target or (f) self-recognize to hybridize with its sequence complement.

resistances of HCAIs necessitate the development of rapid screens that can be reliably employed in hospital settings for both the presence and identification of pathogenic bacteria.

2. BIORECOGNITION ELEMENTS

Many sensors rely on the use of biorecognition elements (Figure 2) to detect pathogens selectively and specifically. The benefits of using these elements are the inherent sensitivity and discrimination they afford, and in this section, we describe the common types of biorecognition elements that are used throughout this review in our discussion of the various sensors, along with the advantages and disadvantages of using each type of element.

2.1. Proteins

2.1.1. Enzymes. Enzymes are catalytic proteins that accelerate chemical reactions, and enzyme binding sites are typically highly selective for their preferred substrates (Figure 2a).[20,21] Enzymatic electrochemical biosensors are some of the most prevalent in point-of-care systems due to the widespread use of the electrochemical glucose sensor in blood sugar monitoring devices.[22,23] Because of their ability both to bind and to convert substrates, enzymes have been applied in the direct electrochemical detection[23] of their target substrates and as a method of signal amplification following binding.[24,25] Both direct enzymatic methods and enzyme-linked methods have been used to detect pathogenic bacteria,[26] and several

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enzymes, including horseradish peroxidase (HRP) and glucose oxidase (GOx), have been ubiquitously applied as signal amplifiers in electrochemical sensors.[24] However, challenges remain in the broad implementation of enzyme-based sensors due to the high cost of protein production and the instability of enzymes, especially following immobilization on a surface.

2.1.2. Receptors. Receptors are proteins that transform chemical signals into cellular activity in a process called "transduction" (Figure 2b). Generally, receptors are divided into three categories based on their activity: they can relay a signal, amplify it, or integrate it. Relaying converts ligand binding to a receptor into an intracellular or intraorganelle signal by altering the confirmation of the receptor to trigger an intracellular response. Amplification involves increasing the effect of a single ligand, generally through the release of multiple signaling molecules for each ligand binding event. Integration involves the incorporation of a ligand binding event into a more complicated biochemical pathway.[20]

Receptor proteins are capable of exceptionally high specificity and sensitivity because they have evolved explicitly to recognize their target chemicals. Electrochemical biosensors based on receptor proteins provide faster and more efficient detection than conventional screening methods, including enzyme-linked immunosorbent assays (ELISA) and quantitative polymerase chain reaction (qPCR). Toll-like receptors (TLRs) are extensively applied to pathogen detection because of their innate recognition abilities.[27] In one example, a complex between the human recombinant toll-like receptor 4 (rhTLR4) and myeloid differentiation-2 (MD-2) immobilized on a gold electrode was used to recognize the endotoxin lipopolysaccharide (LPS).[28] By differential pulse voltammetry, the sensor was confirmed to be specific with a detection limit of 2×10^{-4} EU/mL. Because receptors are capable of both receiving and sending information, they are especially attractive as biosensor components, though no sensors have currently been published that take advantage of both functions of these receptors. However, the expression and purification of these proteins can be exceptionally challenging due to the presence of a hydrophobic transmembrane segment that is usually stabilized by the cell membrane.[20]

2.2. Antibodies

2.2.1. Full Antibodies. Antibodies are glycoproteins that bind their targets with both high specificity and affinity as part of the native immune response to foreign pathogens. They are generally composed of four chains, two heavy chains (V_H) and two light chains (V_L), that form a characteristic "Y" shape. They also have two distinct regions: a crystallizable fragment (F_c) region and an antigen-binding (F_{ab}) region (Figure 2c). The F_c region interacts with other components of the immune system, enabling antibodies to activate an immune response. The F_{ab} region, in contrast, is responsible for recognizing specific target antigens.

One advantage of antibodies for the detection of bacteria is that, for many bacterial pathogens, antibodies already exist because of the natural mammalian immune response to these microbes. Antibodies can also be raised for nearly any target through the injection of inactivated target into an animal host. Targets are often highly complex, meaning that multiple antibodies can be produced against any given target through the activation of multiple B cells. This produces what are known as "polyclonal antibodies".[29,30] In contrast, monoclonal antibodies are produced from a single B cell lineage, which results in antibodies that bind to one specific location, or epitope, of a target.[31,32] Based on the desired antibody application, there are advantages and disadvantages of each type.[33,34]

Most antibodies have especially high affinities for their targets, making them ideal for bacterial detection, and they have therefore been applied in a broad range of assays.[35,36] One of the most prominent techniques involving antibody-based detection is ELISA, discussed in detail in section 3.3.2, in which pairs of antibodies can specifically capture and then detect the presence of a particular target in a 96-well plate format.[37] Because of their ease of use and the abundance of naturally occurring antipathogenic varieties, antibodies have also been used extensively in electrochemical biosensors, especially for the specific capture and detection of bacteria due to the diversity of available antibodies and their specificity for a particular antigen. Antibodies have been applied for the specific detection of *Escherichia coli* O157:H7 and *Salmonella typhimurium* in contaminated food samples through antibody capture of the antigen combined with signal amplification through a GOx-based electrochemical readout. Detection levels of these pathogenic bacteria in food were 10^3 CFU/g.[38] One key advantage of many electrochemical immunosensors is that they generally minimize the amount of required sample preparation, meaning they do not often require preconcentration or enrichment, which can greatly speed up workflow.

2.2.2. Antibody Fragments. Though full antibodies are especially useful, many of their benefits, including specificity and selectivity, can be achieved through the use of only segments of these large proteins, known as antibody fragments (Figure 2c). In fact, antibody fragments can have advantages over full-sized antibodies, often demonstrating reduced nonspecific binding and steric hindrance and, for in vivo applications, reduced native immune responses and improved tissue penetration. There are several common truncations applied to form antibody fragments. The most common is the F_{ab} fragment, also known as the antigen-binding fragment, which consists of one "arm" of the "Y" of the full antibody. The F_{ab} fragment has the advantage of containing only one antigen binding site, which minimizes complications with target binding,[39] and has already proven useful in electrochemical biosensors for pathogen monitoring. Specifically, a biosensor was developed based on the detection of bacterial 16S rRNA through its hybridization to a DNA probe on a surface followed by detection through an HRP-modified F_{ab} against 16S RNA.[40] By varying the sequence of the capture probe on the surface, specificity for a particular bacterial species was attained.

2.2.3. Single-Chain Variable Fragments. An additional class of antibody fragments is single-chain variable fragments (scFv's), which are mimics formed from the fusion of only the variable regions of the light and heavy chains of an antibody (V_H and V_L). scFv's have also been used in electrochemical biosensors, with an scFv that was evolved for the specific capture of *Bacillus anthracis* on a protein chip.[41] An enzymatic tag was added to the antibody fragment of this scFv, and the fragment still maintained its affinity and specificity for *B. anthracis*. In fact, the scFv construct on a chip could detect 1 pg of the protective antigen from *B. anthracis* and less than 100 *B. anthracis* cells in under 2 h.

2.2.4. Monobodies. Though antibodies and F_{ab} fragments are exceptionally useful for the specific capture of pathogenic

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microbes, challenges can arise if no antibody exists for a target of interest because the evolution of new antibodies or fragments is costly and time-consuming. To maintain the specificity advantages of antibodies while easing the complex expression and purification process, antibody mimics have been developed using the fibronectin type III domain (FN3). These alternative scaffolds are quicker and easier to express and purify. This class of antibody mimics maintains a high specificity for biomolecules of interest,[42] and libraries of monobodies can be rapidly generated through variation of a flexible loop region in the FN3 domain.[43,44] Though monobodies have not yet been applied to whole pathogenic bacterial detection, our group has utilized them in an electrochemical sandwich assay for the detection of environmental pollutants in which half of the sandwich was expressed on an *E. coli* surface.[45]

2.3. Oligonucleotides

2.3.1. Aptamers. Because of the challenges associated with the evolution of protein-based biorecognition elements specific to an analyte as well as the issues associated with the stability of these proteins, oligonucleotide aptamers have garnered significant interest. Aptamers have become a staple of biosensors due to their ability to bind to a variety of targets, from small molecules to whole cells. Aptamers are short oligonucleotide sequences (generally less than 80 bases) that adopt a unique three-dimensional conformation that can change upon binding, which enables the aptamers to recognize their target substrate with high fidelity (Figure 2e).[46−48]

2.3.1.1. Selection of aptamers. One key advantage of aptamer technology is the ability to evolve a particular sequence for a desired target rapidly using the systematic evolution of ligands by exponential enrichment (SELEX).[49] One particular class of selection, termed Cell-SELEX,[50] is ideal for identifying nucleic acid sequences that bind bacterial species.[51] Continuing improvements to selection protocols have moved the binding affinity of DNA aptamers from the micromolar to nanomolar range, and these affinities can potentially become even lower with the application of aptamer cocktails (multiple binding sequences used to bind a single target).[52]

2.3.1.2. Bacterial Sensing Applications of Aptamers. Aptamers in hydrogels and on biosensor surfaces have been extensively employed for the capture and detection of pathogenic bacteria, which will later be discussed in terms of impedance-based detection. A colorimetric DNAzyme-based sensor that functioned on a low-cost paper support was developed for the specific detection of *E. coli*.[53] This sensor was capable of discerning between multiple types of bacteria; aptamers against *Acinetobacter baumannii*, vancomycin-resistant *Enterococci* (VRE), *E. coli*, and multidrug-resistant *Staphylococcus aureus* (MRSA) were evolved for immobilization. With this platform, the rapid, low-cost detection of multiple pathogenic bacterial strains was accomplished. An aptamer-based quartz crystal microbalance sensor was also developed for the rapid detection of *E. coli* O157:H7 with a limit of detection (LOD) of 1.5×10^3 CFU/mL.[54] The use of aptamers in biosensors provides more batch reproducibility and stability than the use of antibodies and proteins.

2.3.2. Duplexed DNA. DNA is unique in its ability to self-recognize and therefore specifically hybridize. This feature has led to its broad application in biomaterials[55] and biosensors.[56] DNA hybridization has been extensively used for both identifying bacteria based on RNA sequence and for capturing cells on surfaces.

2.3.2.1. Hybridization for Bacterial RNA Detection. Many biosensors have been based on the specific identification of DNA or RNA hybridization events,[56−59] and for bacterial detection, the monitoring of 16S rRNA (16S rRNA) is a particularly noteworthy example. 16S rRNA is a component of the small subunit of the prokaryotic ribosome. Though most of its sequence is highly conserved, there are several hypervariable regions in the 16S subunit that can be used for the electrochemical detection of multiple species of pathogenic bacteria. Exploiting this detectable variability, an electrochemical sandwich assay was developed for the simultaneous detection of multiple pathogens commonly found in urinary tract infections, including *E. coli*, *Proteus mirabilis*, *Pseudomonas aeruginosa*, *Enterococcus* spp., and several members of the *Klebsiella*−*Enterobacter* group.[60] Capture and detection probe pairs were designed to be species specific. Capture sequences with terminal biotin modifications were immobilized on a multiplexed electrode platform through streptavidin-modified electrodes. The probe sequence was modified with fluorescein to enable the binding of an antifluorescein antibody modified with HRP for electrochemical readout. Amperometric measurements enabled the detection of 2.5×10^3 cells. A similar sandwich assay was developed for the specific identification of *E. coli*, *Enterococcus faecalis*, *P. aeruginosa*, and *P. mirabilis*.[40]

2.3.2.2. DNA Hybridization-Based Cell Adhesion. Bacteria can often be hard to capture for study and detection prior to biofilm formation because of their motility, requiring an alternative scaffold for immobilization. One method to facilitate their specific capture is through DNA hybridization-based cell adhesion. In this method, one sequence of DNA is specifically patterned on the substrate surface, and the bacteria of interest are then modified with the complementary sequence. Upon addition of the cells to the surface, DNA hybridization causes them to adhere specifically in a process termed DNA hybridization-based cell adhesion.[61,62] To modify bacteria with DNA, the cells are initially exposed to sodium periodate, and then hydrazide-modified DNA is added to the cells in the presence of aniline to form a hydrazone linkage between the DNA sequence and oxidized sugars on the surface of the bacteria. This technique has enabled the study of nonadherent cells (such as bacteria prior to biofilm formation) on surfaces.[63]

We have also recently reported the ability to control the density of cells on electroactive surfaces through the surface density of the DNA.[62] Electrochemical DNA quantification combined with fluorescent cell labeling enabled the development of a correlation between the surface density of DNA and surface coverage of multiple types of nonadherent cells. The bound densities of all cell types tested were found to be influenced by the DNA surface density. Mammalian Jurkat and Ramos cells (nonadherent tumor cells) were found to bind at the highest density to low surface densities of DNA. *Saccharomyces cerevisiae* and *Shewanella oneidensis* bound more efficiently to surfaces with higher densities of DNA. This manipulation of DNA surface density presents a new way to control the density of bacterial monolayers.

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3. CONVENTIONAL METHODS OF BACTERIAL DETECTION

A host of commonly used techniques exist for bacterial detection. However, many of these methods have drawbacks that prevent their broad implementation outside of the lab.

3.1. Sample Culture

Culturing bacterial samples is one of the oldest and most prevalent techniques for detecting bacterial pathogens.[64] It relies on bacterial amplification through the growth of a single cell into a colony[65] and is an especially reliable technique that is a staple of microbiology. The main challenge with this method, however, is the time required for many bacteria to be cultured for visible colonies to appear, which can be up to 7 days in the case of *Listeria monocytogenes*.[66] An additional challenge with many bacteria, including *L. monocytogenes*, is that they can become dormant under culture conditions, making growth and bacterial count estimates difficult.[67] Because of the challenges inherent with culturing bacteria, these methods are often used in conjunction with a faster method, such as the polymerase chain reaction (PCR). This has been demonstrated for *Salmonella*,[68] *Listeria*,[69] *E. coli*,[70] and *Campylobacter*.[71,72] The time and labor required for these culture-based bacterial detection strategies have led to the development and popularization of alternative methods.

3.2. Polymerase Chain Reaction Methods

3.2.1. Single Polymerase Chain Reaction Methods. The polymerase chain reaction (PCR) enables the exponential amplification of one or a few copies of a particular sequence of DNA, making PCR-based methods especially powerful because of the small amount of required sample, the speed of the analysis, and the specificity of the method.[73] In fact, PCR amplification is sufficiently sensitive to detect a single bacterial cell,[74] and its extensive application, especially as compared to other bacterial sensors, speaks to its power.

PCR has been used to analyze bacterial contamination in complex solutions. Atlas and co-workers reported the specific detection of coliform bacteria in complex water samples with only 1 fg of genetic material, which corresponds to between only one and five viable bacteria.[75] It is possible to use PCR-based methods to distinguish between living and dead cells, which is not very feasible in binding-based assays. However, this distinction remains difficult with PCR and requires specialized strategies for analysis.[76] Atlas and co-workers again demonstrated the power of PCR-based detection methods with their assay for *Legionella pneumophila*, in which they were able to discern between viable cells (both culturable and not) and nonviable cells based on the DNA sequences that amplified.[77] An additional advantage of PCR is its specificity, even in complex environments. *Vibrio cholerae* O1 was specifically detected by PCR after seeding the bacteria into oysters, shrimp, crab meat, and lettuce. The LODs from this work by Cebula and co-workers were especially impressive, with detection as low as one bacterium per 10 g of food, which is 10 times less than the infection level of this bacterium.[78] Detection of additional bacteria, including *L. monocytogenes*,[79] *Salmonella*,[80] *S. aureus*,[81] and *E. coli*,[82] speaks to the utility of the technique. One limitation of PCR is that only one target can be monitored per experiment.

3.2.2. Multiplexed Polymerase Chain Reaction Methods. Multiplexed PCR maintains all the key advantages of PCR while enabling multiple targets to be monitored simultaneously. This can be especially advantageous for field or point-of-care-based bacterial detection in which more than one bacterial species may be present. Multiplexed PCR has been applied to the simultaneous detection of several types of bacteria, with an assay developed by Kwon and co-workers that detects five different foodborne pathogenic bacteria.[83] Additionally, this method has been extensively applied to the subtyping of single bacterial species, including methicillin-resistant *S. aureus*,[84] *Salmonella* in human stool samples,[85] six types of *E. coli* linked to enteric infections,[86] and enterohemorrhagic *E. coli* in food samples.[87] Because of the speed, ease, and small sample size required for this method of analysis, significant amounts of information can be gained from clinical and field samples in very little time. One key area of improvement for bacterial detection using most conventional PCR methods, both single and multiplexed, involves distinguishing between viable, dormant, and nonviable cells.

3.2.3. Reverse Transcriptase Polymerase Chain Reaction. One variant of PCR, reverse transcriptase PCR (RT-PCR), enables the rapid and sensitive detection of RNA expression through the initial creation of complementary DNA (cDNA) from the target RNA using reverse transcriptase. Subsequent amplification of the cDNA occurs through conventional PCR. RT-PCR is the most sensitive method available to detect mRNA, making it especially useful for the detection of bacteria.[88] One key advantage of RT-PCR as compared to conventional PCR is that amplification of mRNA results in fewer false positives due to the presence of nonviable cells. This was demonstrated by Juneja and co-workers for *L. monocytogenes* through the development of a rapid RT-PCR-based detection method.[89] Additional sensitivity with RT-PCR can be achieved by coupling this method with real-time quantitative PCR (qPCR),[90] which affords an extremely wide dynamic range.

3.3. Immunology-Based Methods

3.3.1. Direct Immunofluorescent Assays. Immunofluorescent assays for bacterial detection rely on fluorophore-labeled antibodies that bind to a bacterial target, with readouts performed generally through either microscopy-based imaging or flow cytometry. This technique has been used extensively for the detection of *Salmonella* in food samples for over half a century[91−93] and was even entered as an official method of analysis for the Association of Official Analytical Chemists.[94] However, its basic application was never found to be sufficiently specific for widespread use, such that more complex immunostaining-based detection strategies have therefore blossomed. For example, Laufs and co-workers developed a double-staining method based on immunofluorescence to distinguish between ingested and extracellular bacteria in human epithelial type 2 (HEp-2) cell culture. This method was able to differentiate between isogenic *Yersinia enterocolitica*, with one strain resistant to phagocytosis while the other was sensitive to it, and differences in internalization were observed.[95] Additionally, immunostaining has been useful for the identification of bacteria in patient samples. Multiple strains of Legionnaires were observed by immunostaining in patient lung samples,[96] and intracellular *E. coli* communities were identified in human urinary tract infections from 100 patient samples.[97] One challenge with many direct immunostaining methods is the necessity of fluorescence microscopy for sample analysis.

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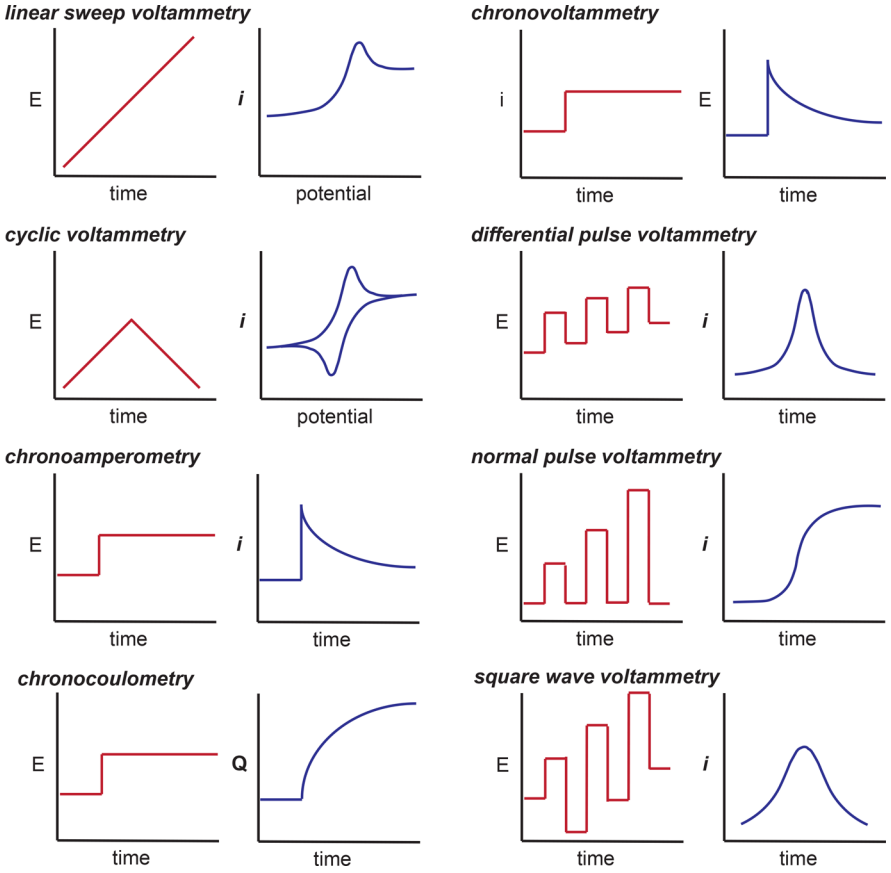


Figure 3. Input/output functions for electrochemical measurements. E = potential, i = current, and Q = charge.

Flow cytometry based methods have additionally been developed to identify bacteria in complex solutions, circumventing the need for a fluorescence microscope in the previously discussed methods while maintaining high reproducibility. Using flow cytometry, Baigent and co-workers developed a sensitive immunofluorescent method to detect *L. monocytogenes* in milk. Immunofluorescence was combined with propidium iodide staining to distinguish *L. monocytogenes* from other foodborne bacterial pathogens, including several species of *Staphylococcus* and *Streptococcus*.[98] Although not widely adopted at the time due to limited access to flow cytometers, flow cytometry based immunofluorescence assays have recently gained popularity. A general flow cytometry method for total bacterial load determination in milk was developed by Veal and co-workers and has an LOD of less than 10 000 bacteria/mL of milk. These results corresponded well with more conventional culture methods.[99] As the cost of flow cytometers continues to decrease, immunofluorescence-based flow cytometry methods for bacterial identification will likely continue to increase in popularity.

3.3.2. Enzyme-Linked Immunosorbent Assay (ELISA). The ELISA is one of the most popular immuno-based assays for the detection of biological agents.[100] In an ELISA, the target is adhered to a surface either specifically or nonspecifically, after which a target-specific detection antibody conjugated to a readout enzyme is added to generate a measurable signal. Although ELISA can be performed using the nonspecific adhesion of the target to a surface, sandwich-based ELISA is more often used for bacterial detection from complex solutions. This type of ELISA involves the initial immobilization of the target through specific capture with an antibody that has been attached to a surface. This can both concentrate the target and reduce false positives because two specific interactions are required for a positive readout.[100]

ELISA has been successfully used to detect foodborne pathogens. An ELISA for *Campylobacter* in bovine samples has been developed and validated with actual bovine samples. *C. fetus* was successfully detected in bovine vaginal mucous samples and preputial washings,[101] and *Salmonella* has been detected in chickens[102] and pigs.[103] ELISA assays have also been extensively applied to the detection of pathogenic bacteria from human samples. A sandwich ELISA was developed for the identification of *E. coli* in patient samples,[104] and *L. monocytogenes* has also been successfully detected by ELISA.[105] Though ELISAs offer highly sensitive and specific detection of target bacteria, they are limited by the many hours they take to complete and the large number of required steps, including washing and blocking in addition to antibody incubation.

3.4. Conclusions

Although conventional methods of bacterial detection and identification have enabled the monitoring of real-world samples, these techniques are all limited by the time required, the necessary instrumentation, which can be costly and highly specialized, and the facilities and personnel required to complete these assays.

4. NONIMPEDANCE ELECTRICAL BIOSENSORS FOR BACTERIAL DETECTION

Electrical biosensors are integrated devices that convert specific biological interactions into electrical signals to provide

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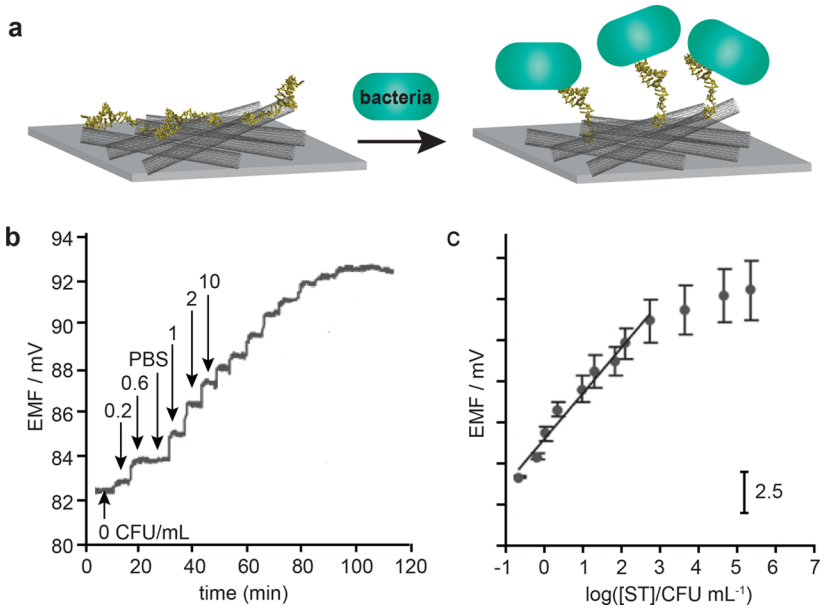


Figure 4. Potentiometric aptasensors to detect bacteria. (a) Aptamer-modified carbon nanotubes are used for the specific capture of the bacteria *Salmonella typhi* (ST). Upon binding of the bacteria to the aptamer, the negatively charged phosphate backbone is removed from the nanotube surface, leading to a potential change. (b) Stepwise increases in potential occur upon addition of increasing concentration of bacteria. (c) Potential changes are a function of the log of the bacterial concentration. Adapted with permission from ref 112. Copyright 2009 John Wiley & Sons.

analytical information.[106] Such sensors are generally thought to be an improvement over more conventional methods for point-of-care or environmental biomolecule detection due to their relative simplicity and low cost.[107] Though the remainder of the review will concentrate on impedimetric biosensors, additional methods of bacterial detection with electrochemistry have proved to be fruitful, and some examples are provided below (Figure 3).

4.1. Voltammetric

Voltammetric sensors are the most prevalent type of biosensor because of their versatility, which stems from the measurement of both potential and current and their ability to analyze a variety of parameters, including voltage (E), current (i), charge (Q), and time (t).[21] Information about the oxidation and reduction potential of the species of interest as well as its concentration can be obtained from this technique. Additionally, multiple redox-active species can be detected simultaneously if their peak potentials are sufficiently separated. A host of voltammetric techniques exist that can be selected based on the desired application.

4.1.1. Varied Potential Techniques. Some of the most common voltammetric techniques measure changes in the current generated as the potential applied to the electrode is varied. The two key techniques are linear sweep voltammetry (LSV), in which the potential is varied at a constant rate as the current output is measured, and cyclic voltammetry (CV), which is similar to LSV though the potential is varied in both the positive and negative directions. Multiple cycles with CV are possible, making it the technique of choice to monitor the stability of products and study reaction intermediates.

4.1.2. Constant Potential and Current Techniques. Two additional voltammetric techniques involve maintaining a constant potential. The first, known as chronoamperometry (CA), monitors the current over time while a similar technique, chronocoulometry (CC), monitors the charge accumulation over time (instead of the current). The final constant input technique often used for biosensors is chronopotentiometry (CP) wherein the current is held constant as the change in the potential is measured.

4.1.3. Pulse Methods. In addition to techniques that involve either holding the potential of a system constant or varying the potential at a constant rate, there are several key techniques that involve pulsing the input potential of a system. Differential pulse voltammetry (DPV), one of the simplest pulse techniques, applies a series of pulses with a defined amplitude (generally between 10 and 100 mV) to scan over a potential window, which removes the charging (nonfaradic) current from the resulting measurement. Current is measured before and after the application of the pulse, and the difference between these measurements is recorded as a function of the base potential. DPV has been elegantly applied to the detection of bacteria for monitoring antibiotic susceptibility. The Kelley group reported a multiplexed device with wells for isolating the bacteria,[108] and then microbial viability was measured by monitoring the conversion of resazurin (RZ) to resorufin (RR) by metabolically active bacteria. If bacteria were susceptible to RZ, no RR would be present. The RZ and RR were differentiated electrochemically by DPV, enabling monitoring of susceptibility and resistance to RZ.

A similar technique is square-wave voltammetry (SWV), in which a square-wave pulse is combined with a staircase potential variation. The net current is reported based on the difference between the forward and reverse steps, centered at the redox potential. Importantly, with SWV, the peak height is directly proportional to the concentration of the electroactive species present, meaning detection limits in the absence of signal amplification can be as low as 10^{-8} M with this method. Furthermore, there is a complete removal of background currents with SWV, and this method is especially rapid. These features enable its application for kinetics measurements and monitoring in tandem with other analytical methods, such as high-performance liquid chromatography (HPLC). Thus,

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SWV provides a more sensitive and often more quantitative measurement than DPV.

4.2. Potentiometric

Potentiometry is one of the oldest instrumental analytical methods, with widespread application due to its low cost and ease of measurement.[109] Potentiometric biosensors monitor the electrical potential of a solution at very high impedance, meaning there is nearly no current flow. Ion-selective electrodes are used to monitor changes in the amount of a charged ion of interest, generally using one of three main types of ion-permeable electrodes: a conventional pH electrode that is permeable to cations; gas-permeable electrodes that measure local pH changes due to the diffusion of gases such as CO_2, NH_3, or H_2S across the membrane; or solid-state electrodes containing a thin layer of conductive material for a specific ion, such as I^-.[110]

Several examples of specific bacterial detection by potentiometry have been developed. One such sensor, involving single-walled carbon nanotubes modified with DNA aptamers, was capable of detecting multiple species of bacteria depending on the sequence of the aptamer immobilized on the carbon nanotubes (Figure 4). These carbon nanotube−aptamer sensors function due to the conformational change of the DNA aptamer upon cell binding. In the absence of analyte, the DNA bases π-stack with the walls of the carbon nanotubes. Upon binding to a cell, the negatively charged phosphate backbone of the aptamer is removed from the carbon nanotube surface, leading to a potentiometric change.[111,112] A sensor specific for *E. coli* was found to have an LOD of 6 CFU/mL, even in complex solutions such as milk.[111] This device was specific for *E. coli* with no cross-reactivity and displayed a linear response to an increase in bacterial concentration up to 10^4 CFU/mL. The carbon nanotube potentiometric biosensor was also demonstrated to be effective for the detection of *Salmonella typhi*.[112] Again, an LOD of sub-10 CFU/mL was observed, with a dynamic range of 4 orders of magnitude. Similarly, a potentiometric aptamer-based sensor on graphene oxide was developed for the specific detection of *S. aureus*[113] that was capable of the impressively sensitive detection of the pathogenic bacteria with an observed LOD of 1 CFU/mL. Though potentiometric biosensors are very sensitive, they continue to be limited by off-target effects and the influence of other components of complex analyte-containing solutions.[109]

4.3. Conductometric

Conductometric biosensors measure changes to the electrical conductivity of a sample solution generated by the progression of an enzymatic or chemical reaction with charged products.[114] These measurements can be particularly advantageous because they are rapid and have a high sensitivity, though obtaining sufficient specificity can be a challenge.[115] One of these sensors has been developed for foodborne pathogens,[116] with a detection system consisting of two components, an immunosensor based on a standard sandwich assay and a signal detection reader, that have been combined into a lateral flow device for enterohemorrhagic *E. coli* O157:H7 and *Salmonella* spp. LODs of this system are approximately 100 CFU/mL within 10 min.

4.4. Capacitive

Capacitive biosensors are based on the electrical double layer theory wherein an electrode coated with a layer of biomolecules that is submerged in an electrolyte-containing solution acts as a capacitor.[117] The capacitance at the surface can be modeled as a series of individual capacitors made up of surface biomolecules[118] wherein each component of the biomolecule layer contributes to the capacitance. When the analyte binds to the surface, the capacitance decreases, enabling sensitive analyte detection.[119] Capacitive biosensors generally contain a layer of small organic molecules to anchor the detection element, which is usually an antibody for specific target recognition. These sensors are exceptionally useful for the very sensitive detection of small molecules, such as endotoxin, which was detected with a 1×10^{-13} M LOD and a linear range of detection from 1×10^{-13} to 1×10^{-10} M.[120]

Importantly, capacitive biosensors have been used to detect whole bacterial cells. An electrode array containing 14 μm × 16 μm pixels could very sensitively detect *S. epidermis* with an LOD of 7 cells/pixel.[121] Microcontact imprinting on electrodes, involving the formation of cavities on the electrode surface specific to the desired bacteria, enabled the detection of *E. coli* using a capacitive electrochemical sensor.[122] In this case, a combination of *N*-methacryloyl-L-histidine methyl ester (MAH) and 2-hydroxyethyl methacrylate (HEMA) were used as monomers, and ethylene glycol dimethacrylate (EGDMA) was applied as a UV cross-linker. The LOD for this platform was found to be 70 CFU/mL, with a linear detection range of 10^2−10^7 CFU/mL. In fact, sufficient interest in capacitive sensors for whole bacterial detection has led to modeling these systems to optimize their LODs.[123] Based on the design rules established through computational methods, a 4-fold improvement over the initial platform design was experimentally confirmed.

4.5. Field Effect Transistor Based

Field effect transistor based biosensors (Bio-FETs) are increasing in popularity for biomarker detection because they generally do not require sample labeling and can be extremely sensitive (Figure 5).[124] Sensing with FETs is based on a change in the source−drain conductivity due to the electrical field in the local environment. Bio-FETs include the addition of a biological recognition element to the transistor. Upon binding of an analyte to the recognition element, the charge distribution—and therefore the surface potential—changes.

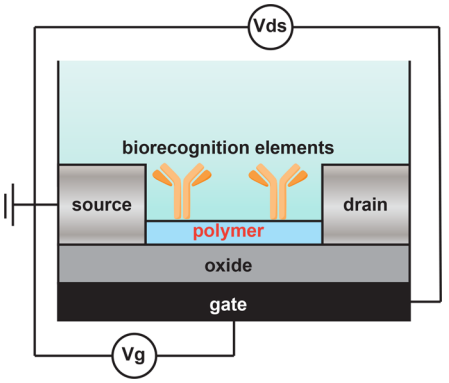


Figure 5. Bio-FET. A bio-field effect transistor (Bio-FET) is based on the same principle as conventional FETs, which respond to changes in the source−drain conductivity. Bio-FETs involve modification with a biorecognition element. Upon binding of an analyte to a recognition element, the surface potential changes.

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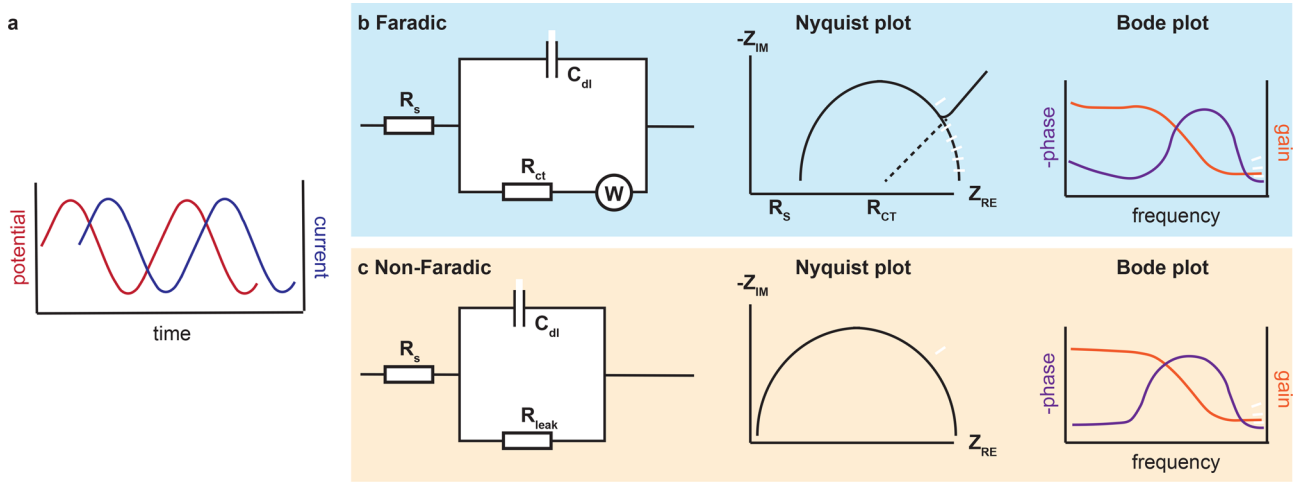


Figure 6. Impedance measurements. (a) Electrochemical impedance is measured by applying an AC potential with a small excitation amplitude to an electrochemical cell and then measuring the current in the cell. (b) Faradic systems involve charge transferred across an interface in the presence of a redox probe. Shown are characteristic models of an equivalent circuit, a Nyquist plot, and a Bode plot for a faradic system. (c) Nonfaradic systems involve assemblies that behave as insulators, and are measured in the absence of a redox probe. Characteristic models of an equivalent circuit, a Nyquist plot, and a Bode plot are shown for a nonfaradic system. R_s = solution resistance; R_{ct} = charge transfer resistance; C_{dl}= double-layer capacitance; R_{leak}= resistance of leakage; W = Warburg element; Z_{RE} = real impedance component; Z_{IM} = imaginary impedance component.

This change leads to a measurable change in conductance between the source and the drain.[125]

Several FET-based sensors have been developed for the detection of bacteria, notably for the specific detection of *E. coli*. Chen and co-workers reported a reduced graphene oxide (rGO) based FET that uses anti-*E. coli* antibodies for the specific capture and detection of these pathogenic bacteria.[126] Only 1 μL of sample was required for detection with their sensor, which could directly detect down to 10^3 CFU/mL of *E. coli* with an extrapolated LOD of one single bacterial cell. Their sensor was additionally capable of detecting 10^4 CFU/mL of *E. coli* in the complex matrix of river water, demonstrating its utility for real-world applications. Similarly, Estrela and co-workers developed a metal oxide semiconductor field-effect transistor (MOSFET) based sensor for detecting the strain of *E. coli* most often responsible for UTIs (uropathogenic *E. coli* or UPEC).[127] By utilizing α-D-mannose-modified surfaces as a biorecognition element, the researchers could specifically capture UPEC that express type 1 fimbriae, which are attachment pili that bind to mannose. These pili enable the specific capture of UPEC on the α-D-mannose-modified surfaces. With detection in under 2 s, the limit of quantification (LOQ) with this device was 2×10^5 CFU/mL. A graphene FET has also been developed for the specific capture of *E. coli*. In this case, a DNA aptamer modified with a pyrene was used as a specific biorecognition element for the bacteria,[128] and the binding of negatively charged cells increased the density of holes in the graphene, which were pumped into the channel through the source–drain electrodes. The LODs were found to be 100 CFU/mL.

In addition to *E. coli* detection, several groups have developed selective FET sensors for additional pathogenic targets. Maxwell and co-workers reported the development of a complementary metal oxide semiconductor (CMOS) based FET for the detection of bacteria using a generalizable bacteriocin-based strategy.[129] Bacteriocins are proteins produced by one bacterial strain that are active against closely related strains. Upon bacteriocin interaction with a specific type of bacteria, potassium is released from cells that can be measured using a potassium-selective FET, enabling especially low detection limits (10^7 cells/mL). Generally, issues in mass producing bio-FETs limit their widespread implementation as commercial biosensors.[130]

5. IMPEDANCE ELECTROCHEMISTRY THEORY

Fundamentally, impedance is the effective resistance of an electrical circuit to a component in response to an alternating current (AC) (Figure 6a). Impedance-based techniques are some of the most reliable and reproducible characterization methods for electrochemical systems, and information regarding double layer capacitance, diffusive impedance, and charge transfer resistance characteristics can be easily extrapolated.[131] Impedance is useful for characterizing systems including semiconductors, batteries, fuel cells, and biosensors. Basic impedance (Z) measurements are made by applying small voltage perturbations to a system and monitoring the corresponding current response. Thus, impedance can be expressed simply as the quotient of the voltage–time function and the current–time function. The current–voltage ratio is measured using the electrochemical impedance spectroscopy (EIS) technique, in which the frequency of the applied sinusoidal voltage is varied.[132]

5.1. Expressing and Visualizing Impedance Data

Because impedance values are complex numbers, there are two common ways to express impedance data (Figure 6). The first involves the coefficient of the absolute value of Z ($|Z|$) and the phase shift (ϕ). These values can be visually represented as a Bode plot, where $\log(|Z|)$ and ϕ are plotted as functions of $\log(f)$, where f is the frequency of the AC current. The alternative method of visually representing data is with a Nyquist plot, in which the real component of the impedance (Z_{RE}) is plotted against the imaginary component (Z_{IM}) at each frequency.[133]

5.2. Methods for Measuring Impedance

Based on the desired application of a biosensor, several different methods for measuring the impedance may be used.

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5.2.1. Single-Frequency Measurements. Single-frequency measurements are the simplest for measuring impedance and are generally limited to impedance microbiology (IM) monitoring applications. As an example, measuring conductance changes at a particular frequency allows for easy monitoring of bacterial growth.[134]

5.2.2. Impedance-Splitting Methods. Impedance-splitting methods are an extension of single-frequency measurements and are also generally limited to IM. With these techniques, the "total impedance" can be obtained by measuring the impedance at two frequencies. One frequency, generally under 100 Hz, enables the determination of interface impedance at the electrode surface, and the second frequency is chosen (about 10 kHz) to provide the electrolyte impedance from the solution.[135]

5.2.3. Electrochemical Impedance Spectroscopy (EIS). Electrochemical impedance spectroscopy (EIS), measuring the current as a function of an applied sinusoidal voltage of a given frequency, is one of the most common methods of impedimetric analysis in biosensors because of the amount of information that can be obtained. The measurement can be repeated over a variety of frequencies, meaning that the resulting data can be used to elucidate many electrochemical phenomena.[136] The data from EIS measurements can be plotted in multiple ways to extrapolate a variety of parameters pertaining to the system.

5.3. Equivalent Circuits

To extrapolate relevant parameters from EIS data, the raw impedance data can be fitted to a model system (Figure 6). Such systems are modeled as an equivalent circuit, with the Randles and Ershler model the most common for EIS analysis.[137−139] This model is mostly used to obtain information about interfacial phenomena, including the electron transfer resistance at a surface (R_{et}), the double-layer capacitance at the surface (C_{dl}), the diffusive resistance in solution (R_s), and the Warburg diffusion element (Z_W), which models linear diffusion to a surface, such as a planar electrode.

5.4. Types of Impedance Sensors

Impedance sensors can be divided into two families based on their surface characteristics and how measurements are taken. The differences between the two types of sensors are defined by their charge transfer capability characteristics, which can be either faradic or capacitive.[140]

5.4.1. Faradic. Generally, electrochemical faradic systems are defined as those in which charge is transferred across an interface (Figure 6b). Faradic impedance systems involve electrode surfaces that are partially or completely covered by a noninsulating layer or an insulating layer capable of catalyzing redox transformations. Faradic EIS sensors generally involve the incorporation of a redox-active moiety into a solution that is alternately reduced and oxidized via the electrode.[132] To prevent depletion of the probe molecule, an overall DC bias is maintained throughout the measurement. Measured changes in these systems are generally reported based on changes to the charge transfer resistance (R_{ct}). Generally, such sensors involve surface binding events that block a portion of the surface and increase the R_{ct} value.[141]

5.4.2. Nonfaradic/Capacitive. Capacitive sensors involve surfaces that are fully covered by a dielectric layer, meaning that the complete assembly behaves as an insulator and no redox probe or label is required or present with this type of sensor (Figure 6c). Generally, interactions or binding events at an electrode surface decrease the double-layer capacitance (C_{dl}).[141] Capacitive sensors are generally monitored at a single frequency.

6. IMPEDANCE PROPERTIES OF CELLS

The complex nature of bacterial cells means that their electrochemical properties are also highly complicated. Cell membranes are comprised of a lipid bilayer, with the lipids oriented such that the polar head groups face the aqueous environment, on both the interior and exterior of the cell, and the hydrophobic hydrocarbons of the lipid tail form the interior of the membrane. This yields a membrane that is highly insulating,[134] estimated to be in the range of 10^{-7} S/m.[142] The bilayer also has many embedded proteins, some of which are ion channels that can be treated as resistors, such that the resistance across the full membrane can be modeled as "parallel ion channel resistors". Estimates of the total resistance across cell membranes vary from 10^5 to 1 M $\Omega\cdot\mu m^2$.[143,144] The interior of bacterial cells also contains a variety of charged biomolecules and small molecules, rendering it highly conductive (as high as 1 S/m).[142] Because of the insulating properties of cell membranes, if cells bind to an electrode surface, they reduce the electrode area that is accessible to redox-active molecules in solution, leading to an increase in the interface impedance.

7. IMPEDANCE MICROBIOLOGY

Impedance microbiology (IM), or the monitoring of bacterial growth density by monitoring the electrical parameters of the growth medium, is one of the earliest methods of bacterial detection developed with impedance electrochemistry. It has been used as such for over a century,[145] with early IM monitoring systems composed of two planar electrodes submerged in solution. Such platforms have found extensive use in the development of point-of-care systems.[146]

Bacteria can be detected using IM through either a direct or indirect measurement, which is chosen based on the ionic strength of the solution to be measured. The direct detection method involves monitoring the AC impedance at a single frequency at a pair of electrodes immersed in bacterial culture medium (Figure 7).[147,148] Impedance changes result from ionic metabolites secreted by the bacteria into the solution. If impedance changes are detected beyond a threshold level, the sample is positive for bacteria.

Indirect detection may be required if the ionic strength of the solution is too high to detect clear differences due to bacterial ion secretions.[149] In this case, CO_2 is instead monitored in a chamber that is isolated from the growth chamber. The rapid speed and ease of use of this technique have led to its broad application in bodily fluid sampling and foodborne pathogen detection.[134,150]

7.1. Applications in Healthcare

Because this method of bacterial detection does not require pure or optically transparent samples, it has been used for determining bacterial load in bodily fluids for nearly five decades. Early reports of the application of the Bactometer for impedance-based bacterial detection in urine demonstrated a nearly 97% agreement between impedance and conventional measurements in patient urine samples.[151] Similar methods were used to identify bacteremia in patient blood samples that were lysed, filtered, and measured for bacterial count through impedance as well as conventional culture methods.[152] It was

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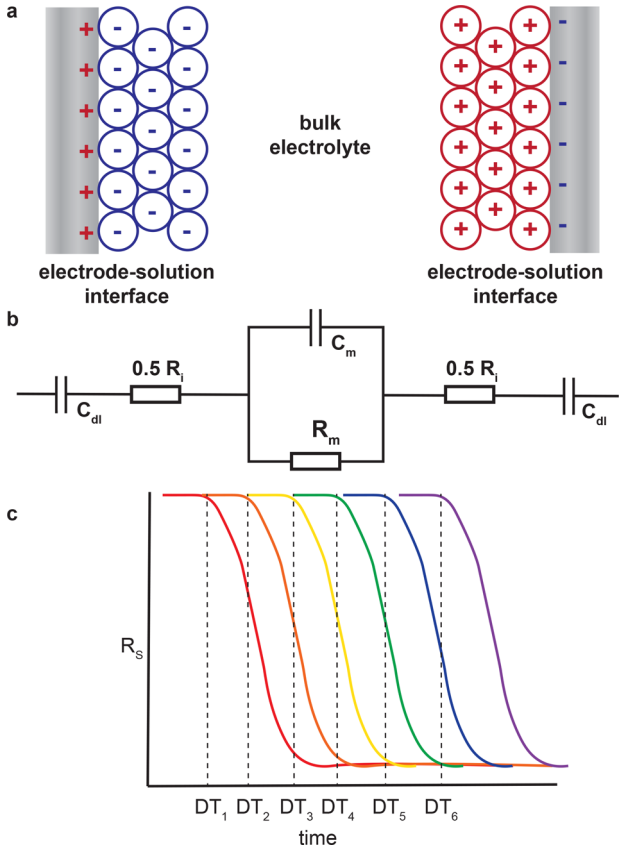


Figure 7. Impedance microbiology. (a) Impedance microbiology is based on monitoring of AC impedance at a single frequency using a pair of electrodes immersed in a bacterial culture medium. Upon microbial metabolism, impedance changes are detected due to conductivity changes in the bulk electrolyte. (b) A example equivalent circuit is shown for a pair of electrodes in contact with a bulk electrolyte medium. (c) Idealized R_s curves are shown for bacterial growth in solution over time. R_s = solution resistance; R_m = resistance of the medium; C_m = capacitance of the medium; R_i = resistance of electrode−electrolyte interface; C_i = capacitance of the electrode−electrolyte interface; DT = detect time.

found that the impedance-based measurement techniques were able to identify 36% more infections than the conventional methods. Impedance monitoring was then expanded to the detection of nine pathogens commonly found in blood cultures, including *S. aureus* and *E. coli*.[153]

7.2. Applications in Food Safety

Beginning in the late 1970s, IM was suggested as a rapid alternative to plating and counting colonies for the evaluation of the microbial burden in milk,[154] and it has been used for pathogen detection in food ever since. In fact, IM is a certified method for the detection of viable cells, namely *Salmonella*, by the Association of Official Agricultural Chemists.[155] Few improvements were made to the analytical methods used for agricultural applications between the late 1970s and the early 2000s, though many examples of IM are found in the literature during this time. Many applications involve the use of commercial measurement tools: the Bactometer[156−158] and the Malthus systems.[156,159] Improvements in IM detection include moving to interface capacitance and conductance,[135] an alternative that was found to be advantageous over even conventional conductometric systems. IM has also been applied to bacterial detection from meat,[160] fish,[161] and contaminated water.[162]

One interesting variation on conventional IM was provided by Lorenzelli and co-workers.[155] They quantified solution concentrations of *E. coli* for foodborne pathogen detection using bacteriophage-mediated cell lysis. In a set of proof-of-principle experiments, the researchers designed a low-volume (nanoliters) flow cell and low conductivity broth for detection. A linear range of detection was obtained between 10^4 and 10^8 CFU/mL, with an LOD of 1 CFU/well on the device. Such improvements to IM will continue to increase the applicability of this method.

7.3. Impedance-Splitting Methods

Though conventional IM methods generally monitor changes in solution conductance at a single frequency, it has been found that effective bacterial concentration measurements using these techniques should include a measurement of the interface impedance at the electrode surface.[163] As described in section 5.2.2, differentiating these components can be accomplished through impedance measurements at multiple frequencies, and "impedance-splitting" methods that automatically do this have been developed for bacterial detection.

7.3.1. BacTrac System. Because of the large amount of information that can be obtained about both the bacteria and their produced metabolites using impedance-splitting methods, BacTrac, a commercial sensor that measures dual frequencies, has been developed.[164] BacTrac has been used extensively to monitor the bacterial load in food products. This system operates by taking initial measurements of the electrode impedance (Z_E) and the culture medium impedance (Z_M) and monitoring the change in each of these parameters over time. The ability to monitor both values can be useful for media with an especially high ionic strength that interferes with conventional IM measurements. Based on the type of microbe and its respiration rate, either one or both parameters can be used with this system.

7.3.2. Applications of Impedance Splitting for Foodborne Pathogen Detection. With the ease of use of the BacTrac system, impedance-splitting methods have been applied to the detection of a variety of foodborne pathogens from many different sources. One example demonstrated standard measurements for both the Z_E and the Z_M for various species of bacteria, including several strains of *Salmonella* and *Enterobacter*.[165] Following initial characterization of the growth of these strains using the BacTrac system, measurements from skin washings of newly slaughtered poultry were taken. Based on several misidentifications and false positives, the authors concluded that their system was suitable for use as a negative control but not for the specific identification of bacterial strains.

In a similar study with a BacTrac system, levels of *Bacillus stearothermophilus*, one of the bacteria most often cited as a source of food spoilage, were measured.[166] For test measurements in media, it was found that the Z_E parameter changed significantly in the presence of bacteria while the Z_M changed little. Based on the observed changes to the Z_E values upon bacterial growth, both planktonic and adhered cells could be detected in both vegetative and spore forms.

Importantly, studies have also compared detection using the BacTrac system to conventional colony growth on plates to validate this technique. *Yersinia enterocolitica* was detected in tryptic soy broth using impedance-splitting methods,[167] and

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the results were compared to those obtained for colony growth on plates, confirming the validity of the significantly faster impedance-based method. Calibration curves for each parameter (Z_E and Z_M) were generated for several strains of *Y. enterocolitica*. This enabled surface CFU measurements much more rapidly than counting colonies on plates, and the methods were in good agreement for the number of CFUs in samples with unknown concentrations.

With validation and broad applications in foodborne pathogen detection, examples of impedance-splitting bacterial detection in real contaminated foods also exist. One such example is the direct detection of *Alicyclobacillus acidoterrestris*, a thermostable bacterium often found to cause spoilage in fruit juices.[168] It is challenging to detect this species of bacteria using conventional methods, as it does not generate a gas upon spoilage but instead generates guaiacol, an aromatic oil. Fruit juice samples were initially heat shocked to activate the *A. acidoterrestris* spores prior to detection. Because of the versatility of the BacTrac system, both direct and indirect impedance measurements were obtained. The indirect detection strategy proved optimal for the sensitive detection of this contaminant in fruit juice samples with too high of an ionic strength for conventional IM measurements.

8. INTERDIGITATED ELECTRODES FOR NONSPECIFIC DETECTION

Improvements to electrode platforms have enabled increased detection sensitivity and speed without requiring specific biomolecular interactions. The second generation of IM devices moved away from macroelectrodes to more sophisticated interdigitated electrode arrays (IDAs), which are generally composed of two individually addressable electrode strips containing multiple microelectrodes (Figure 8). Each set

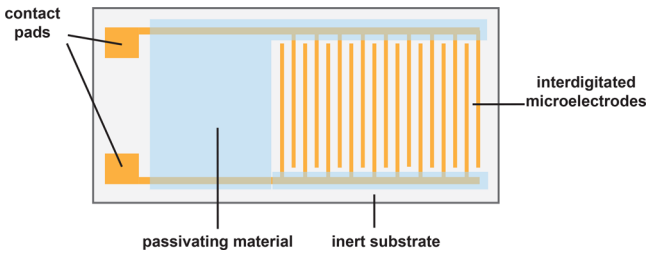


Figure 8. Interdigitated electrode array (IDA). IDAs contain two individually addressable electrode strips comprising multiple microelectrodes. Each set of microelectrodes acts as a pole for bipolar impedance measurements.

of microelectrodes can therefore act as a pole for bipolar impedance measurements. These electrodes maintain the benefits of conventional electrochemical biosensors while improving detection limits by optimizing the size and spacing of these electrodes. Such microelectrodes provide significant advantages over conventional ones, including high signal-to-noise ratios, low resistances, small required detection volumes, and rapid equilibration, even in the absence of modifications for specificity.[134]

8.1. Detection of Metabolic Activity of Bacteria

As with macroscopic electrodes for IM-based bacterial detection, microelectrodes can also detect bacteria either directly or based on the metabolites produced. Impedance monitoring of metabolic activity also ensures that the detected bacteria are viable. An IDA-based chip was developed to monitor *L. monocytogenes*, typically responsible for the listeriosis family of diseases. An initial version of the platform that could detect metabolic differences between living and dead cells using IM was developed using the innocuous *L. innocua*.[169] Platform optimizations reduced the working volume of the wells to 5.27 nL.[170] It was then validated with three bacterial species: the pathogenic *L. monocytogenes*, the innocuous *L. innocua*, and nonpathogenic *E. coli*. Changes in impedance were found to be correlated to metabolic activity, with little to no change observed for heat-killed samples. The LOD for *L. monocytogenes* was found to be 200 cells, which is on the same order of magnitude as the infectivity of this bacterium. The improved sensitivity and decreased sample volume of these platforms have enabled the sensitive metabolic monitoring of living pathogenic bacteria.

Similarly, a microelectrode array was developed by Bahreyni and co-workers to monitor the number of bacterial colonies growing on the device using changes in the impedance of the media.[171] As compared to existing colony counting technologies, this device accelerated the detection of foodborne pathogens. A single CFU of *E. coli* was measurable following 16 h of incubation on the electrode array. A speedier device was later developed by Ebrahimi and co-workers using three-dimensional (3D) gold−nickel electrodes that incorporated a gel-based medium for in situ detection without sample preparation.[172] To construct their device, the initial fabrication of the interdigitated electrode array was followed by a coat of deionized water agar. Finally, a layer of nutrient agar was cast on top of the water agar. Impressively, the metabolic activity of 10^4−10^6 cells/mL from 1 μL was detected after only 1 h. This limit of detection is on the order of what is required for clinical diagnosis of a UTI.

8.2. Direct Detection of Bacterial Cells

Because of their high signal-to-noise ratios and low required volumes, IDA-based IM sensors have been applied in a variety of areas ranging from the detection of foodborne pathogens to monitoring biofilm formation.

8.2.1. Bare Interdigitated Arrays. An IDA-based platform was developed to monitor the growth of *S. typhimurium* in both media and milk.[173] It was optimized for monitoring bacteria in complex media, and optimizations included the development of a new equivalent circuit for data fitting comprised of a dielectric and a double-layer capacitor, as well as a resistor for the media. The LOD of this platform was 4.8 CFU/mL, with measurable changes to impedance occurring between 10^5 and 10^6 CFU/mL, even in complex solutions such as milk. With this platform, specificity was obtained based on the choice of media.

Unmodified IDAs have also been extensively used to measure bacterial growth and biofilm formation. As an example, an IDA platform was developed and optimized to monitor biofilm formation from *P. aeruginosa* PAO1 cells, with a full EIS performed from 1 Hz to 100 kHz.[174] It was found that the double-layer capacitance significantly decreased in the presence of bacteria for 1 h. Importantly, fixed-frequency impedance was found to be applicable for the detection of initial bacterial attachment to the IDAs, which increases the number of available interfaces that can be used for measurement.

Such direct techniques for detecting biofilm growth have also been multiplexed to be compatible with conventional 96-

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well plates. Arana and co-workers developed an IDA assay with EIS readout to monitor the growth of various *Staphylococcus* species over 20 h of growth.[175] They were able to monitor two *S. aureus* and two *S. epidermidis* strains over the 20 h time course. Low measurement frequencies were found to be optimal for the detection of biofilm formation on their IDAs. Such multiplexed platforms will be invaluable for the continued monitoring of infectious bacteria for healthcare applications.

Similar methods were used to monitor *P. aeruginosa* PA14 biofilm formation in a 96-well plate using single-frequency impedance measurements over 72 h.[176] After 35 h, changes to the slope of the curve correlated to the stage of biofilm formation. Impairment in biofilm formation was monitored by this method and compared to crystal violet staining and confocal microscopy. As the impedance results agreed with traditional biofilm-monitoring staining methods, the authors posit that this impedance method is viable for both characterizing biofilm formation and monitoring small molecule treatments to inhibit biofilm formation.

8.2.2. Peptide-Modified Interdigitated Arrays. Though bare IDAs can sensitively monitor and detect many bacterial species, the peptide modification of electrode arrays has provided more biological information than bare IDAs alone.[177] McAlpine and co-workers developed a portable biosensor for pathogenic bacteria based on IDAs modified with antimicrobial peptides (AMPs).[178] By modifying gold electrode arrays with magainin I, a semiselective AMP, through a terminal cysteine, *E. coli* were captured at clinically relevant levels of 1 cell/μL. Both *E. coli* and *Salmonella* were detected, and upon incorporation of the electrode array into a microfluidic device, *E. coli* were detected in real time. Based on the bacterial-capture ability of AMPs, the McAlpine group also developed a wireless sensor to monitor bacteria on teeth.[179] Graphene nanoelectronics, along with thin-film inductor−capacitor (LC) resonant circuits, were directly integrated onto silk thin films, enabling direct interaction with tooth enamel, battery-free operation, and wireless readout. The graphene was modified with the AMP odorranin-HP, which has demonstrated activity against *E. coli*, *H. pylori*, and *S. aureus*. The detection limit of the device was found to be 100 CFU/mL of *H. pylori*.

A similar strategy was employed by Gil and co-workers on their three-dimensional interdigitated electrode array (3D-IDEA) platform.[180] By immobilizing a synthetic AMP (hLf1-11) on the 3D-IDEA device, the sensor was found to rapidly detect the periodontopathogenic *Streptococcus sanguinis*. Within 1 h, the device could detect 10 CFU/mL of *S. sanguinis* in KCl solution and 100 CFU/mL in artificial saliva. This device is promising for monitoring infection following periodontal implants.

8.3. Direct Bacterial Detection with Antibody-Based Preconcentration

Using IDAs and IM-based detection, an extension of the IDA sensor for *S. typhimurium* was developed that could successfully measure samples in brain heart infusion (BHI) broth.[181] Specificity was achieved by initially treating the samples with anti-*Salmonella*-coated magnetic beads. A linear detection range of 10−10^6 CFU/mL was determined for this platform. The ease of use and low detection limit of these platforms makes them attractive for the field monitoring of foodborne illnesses.

A similar IDA platform was developed for the detection of *E. coli* O157:H7 from ground beef samples that were preconcentrated with anti-*E. coli* beads that functioned to preconcentrate samples as well as control their placement within the IDA platform.[182] This resulted in sensitivities of 8.0×10^5 CFU/mL for detecting *E. coli* from the ground beef samples. Such preconcentration steps decrease the time required for detection and improve the sensitivity of the platform.

In an additional example, Li and co-workers developed a strategy for the initial capture of either *E. coli* O157:H7 or *S. typhimurium* with antibody-labeled magnetic beads.[38] Secondary labeling was subsequently performed with GOx-modified antibodies. The magnetic bead−GOx modified cells were added to a low ionic strength medium containing glucose. The GOx then oxidized the glucose, increasing the ionic strength of the solution and therefore decreasing the impedance at unmodified gold IDAs. The dynamic range of the sensor for both types of bacteria was 10^2−10^6 CFU/mL. From food samples, the LODs were exceptionally low, with an LOD for *E. coli* in ground beef of 2.05×10^3 CFU/g and an LOD for *S. typhimurium* of 1.04×10^3 CFU/mL.

8.4. Microfluidics Coupled with IDA for IM Detection

In addition to the advantages that IDAs offer in terms of a small sample size and high signal-to-noise ratio, combining microfluidics with IDAs for IM-based detection has facilitated the incorporation of both in-device preconcentration methods and the combination of multiple IDAs to increase sensitivity. Bashir and co-workers reported improvements to IDAs on chips through the incorporation of a dielectrophoretic preconcentration step prior to IM bacterial detection, which afforded orders of magnitude more concentrated samples (10^4−10^5-fold more concentrated).[183] This platform enabled rapid, sensitive detection of *L. monocytogenes* from a final sample volume of 400 pL and provided a 7-fold decrease in the time required for sample detection with the preconcentration step, making it an important consideration in future platform designs.

Preconcentration was similarly accomplished through droplet evaporation prior to detection. As reported by Alam and co-workers, the inherent osmoregulatory pathways in bacteria can be triggered by droplet evaporation on an electrode.[184] By monitoring changes in conductance in the droplet due to the triggered osmoregulatory pathways in bacteria present in the sample, the bacteria were detected in minutes. Because the assay was not dependent on bacterial growth or doubling time, the detection time was found to be species independent. The differentiation between both wild type and genetically engineered *S. typhimurium* and living versus dead *E. coli* and *S. epidermidis* was accomplished. Detection down to 10^4 living cells/mL was reported.

In addition to the speed and sensitivity improvements afforded by the incorporation of a dielectrophoretic preconcentration step, additional sensitivity in IDA-based devices has been achieved through the incorporation of fluidics[185] and multiple IDAs in a single sample well.[186] For IDA platforms combined with fluidics, LODs were found to be 1.6×10^2 CFU/mL in pure cultures and 1.2×10^3 CFU/mL from raw beef samples.[185] To obtain improved sensitivity of IM-based monitoring, sample chambers of only 60 nL were incorporated into the platform, and double IDAs were then developed in combination with *E. coli* O157:H7 preconcentration in their growth medium (Figure 9). The detection range for this improved platform was between 8.0 and 8.2×10^8 CFU/mL,[186] significantly extending the range of detection over

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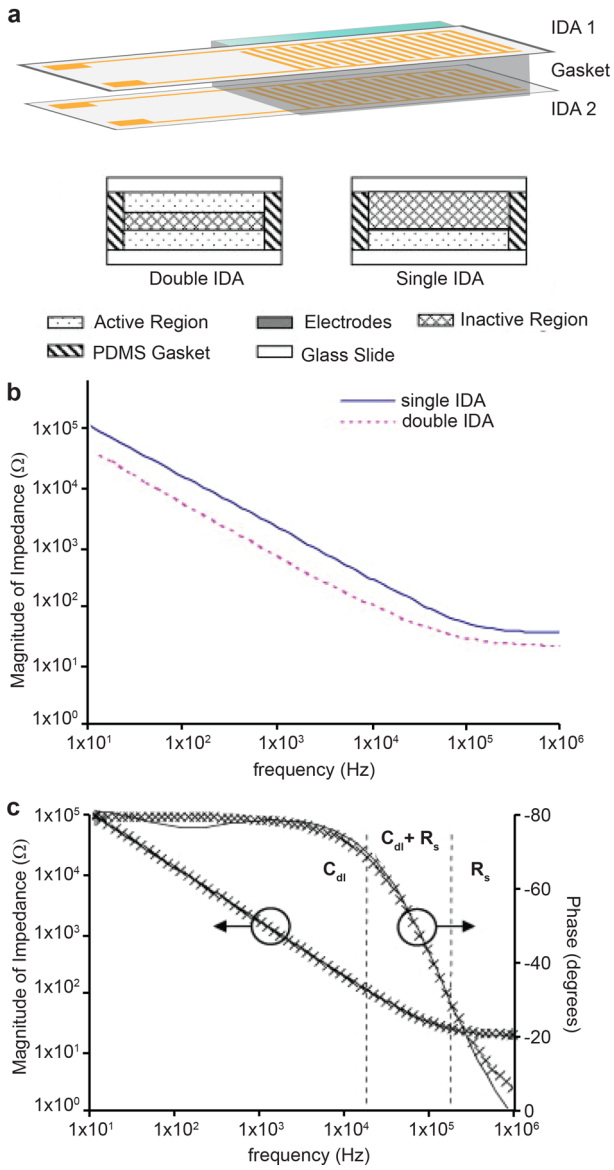


Figure 9. Double interdigitated electrode array (IDA) platform for *E. coli* detection. (a) A double IDA configuration is shown, with one IDA above the well containing bacteria and one below. The side-on view of the setup emphasizes the increase in the active detection region upon incorporation of a secondary IDA. (b) A direct comparison is shown for single IDA and double IDA platforms for 8.2×10^4 CFU/mL *E. coli*. (c) A Bode plot of experimental data and curve-fitted plots are shown for the equivalent circuit for *E. coli* detection in YPLT medium. R_s = solution resistance; C_{dl} = double-layer capacitance. Adapted with pennission from ref 186. Copyright 2008 Elsevier.

the previous generation of the platform. The improved sensitivity was attributed to the increased electrode-detection surface area.

8.5. Signal Amplification with IDAs

Though the preconcentration of bacteria can be useful for improving LODs for IDA bacterial sensors, even this may be insufficient for the sensitive detection and, importantly, quantification of cells. In many cases, signal amplification methods can greatly improve LODs and linear detection ranges. There are a variety of strategies for signal amplification, ranging from enzymatic methods[66] to those involving nanoparticle-coated cells.[187]

Certain enzymes, namely HRP, GOx, alkaline phosphatase, and β-galactosidase, are often incorporated into electrochemical platforms because of their signal amplification abilities through catalytic substrate turnover.[66] An additional strategy for signal amplification is through the application of nanoparticles in combination with silver coating, the foundation of which can be found in work on DNA detection. The Mirkin group initially reported an electrochemical strategy for detecting single-stranded DNA (ssDNA) through the immobilization of gold nanoparticles (AuNPs) followed by AuNP-catalyzed silver precipitation.[188] A similar method was reported by Li and co-workers on graphene, in which ssDNA adhered to the graphene through π-stacking interactions.[189] The DNA was then coated in silver through AuNP-catalyzed precipitation. Wang and co-workers were inspired by this work to develop a method of signal amplification for bacterial detection.[187] A microfluidic device was constructed that contained an initial well for mixing with the bacteria. The cells, specifically *E. coli*, were coated in a positively charged polymer (diallyldimethylammonium chloride [PDDA]), followed by a coating of AuNPs. Finally, AuNP-catalyzed silver adducts were formed on the bacterial surface, which could then be concentrated by dielectrophoresis (DEP), and the increased conductance of the solution and decreased impedance were measurable. The linear detection range of the platform was 2×10^3–2×10^5 CFU/mL with an LOD of 500 CFU/mL.

8.6. Interfacing IDAs for Real-World Detection

Efficient preconcentration methods have also been combined with unique interfaces for the rapid detection of bacteria in real-world situations. One of the key challenges that remains with impedimetric detection is the burden of interfacing the device with a potentiostat. To circumvent this challenge and bring IM-based IDA platforms to a broader audience, Liu and co-workers developed a disposable platform for the preconcentration and detection of bacteria that wirelessly interfaced with a smartphone to enable rapid and sensitive detection without requiring additional instrumentation.[190] They found that preconcentrating the samples was essential for achieving the exceptionally low LOD of 10 bacteria/mL. Such platforms will facilitate the rapid quantification of bacteria in field water samples, especially drinking water.

Overall, IDEs have shown improved detection with IM as compared to bulk electrode measurements. Further improvements to detection schemes can be made through the incorporation of preconcentration steps and microfluidic components.[191]

9. IMPEDIMETRIC IMMUNOSENSORS

Impedimetric immunosensors are a specific class of impedimetric detectors that monitor the interaction between antibodies and their antigen targets on a surface.[192] This class of sensors is one of the most prevalent because of the inherent specificity provided by the antibody component. Impedimetric immunosensors for bacterial cell detection are generally constructed by immobilizing specific antibodies to target bacteria on an electrode. The binding of bacterial cells to the surface changes the electrical properties of the electrode surface, either due to the inherent properties of cell membranes or by blocking diffusing redox-active molecules from interacting with the surface. In the absence of a redox

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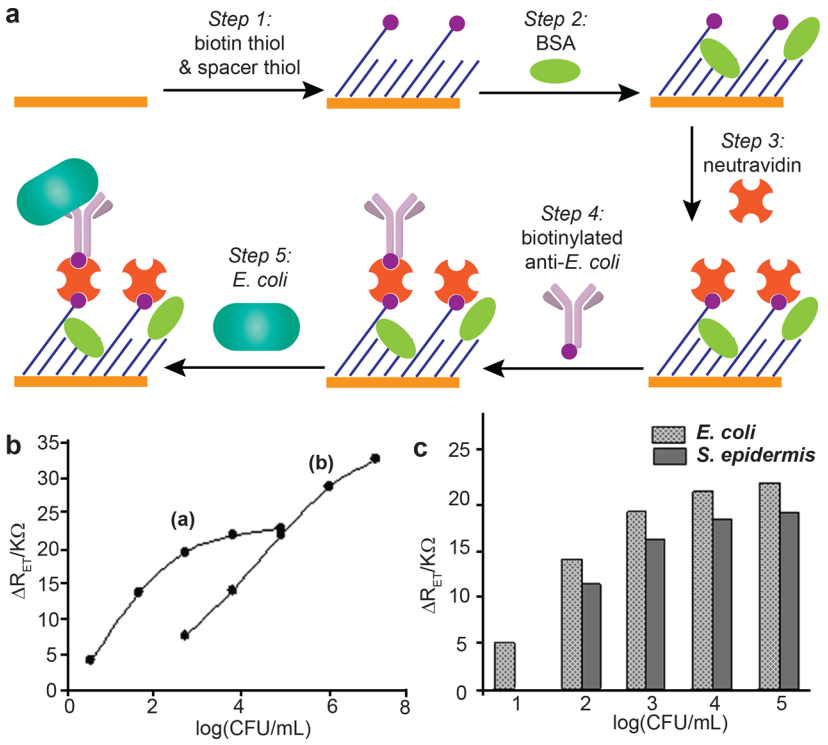


Figure 10. Biotin–neutravidin-based impedimetric immunosensors. (a) Stepwise assembly occurs via neutravidin-based immobilization of biotin-labeled antibodies on electrode surfaces. (b) Detection of bacteria is based on the change in R_{ET} (electron transfer resistance) for (a) whole bacteria and (b) lysed bacteria as a function of the log of the bacterial concentration. (c) The sensor is specific for *E. coli* and *S. epidermidis* based on the impedance detection signal. Adapted from ref 198. Copyright 2007 American Chemical Society.

probe, the impedance signal is a direct result of intact bacteria and is influenced by the morphology, number, and growth of the bacteria. In contrast, if the impedance is measured in the presence of a redox probe, changes in faradic impedance are instead measured and are generally the more prevalent technique for monitoring specific binding events to modified electrodes.[134]

9.1. Nonspecific Antibody Adsorption

The easiest method for modifying a surface with antibodies is through nonspecific adsorption. Antibodies, like many polypeptides, naturally adsorb onto a variety of substrates without requiring covalent chemical connections. This modification method lacks control over the conformation of biomolecules on the surface but is the easiest method.

9.1.1. Indium Tin Oxide (ITO) Electrodes. Antibodies for the detection of *E. coli* O157:H7 were adhered to an IDA of ITO electrodes on glass through nonspecific adhesion.[193] Using EIS in the presence of $Fe(CN)_6^{3-/4-}$, the charge transfer resistance from the surface (R_{CT}) was monitored upon *E. coli* binding. A linear detection range of 4.36×10^5–4.36×10^8 CFU/mL was obtained with this platform.

9.1.2. Glassy Carbon Electrodes. An additional example of antibody adhesion for the formation of an impedimetric immunosensor is the work of Jiao and co-workers.[194] They fabricated an impedance biosensor for the detection of *Campylobacter jejuni* in diarrheal samples from immobilized *O*-carboxymethylchitosan surface-modified Fe_3O_4 nanoparticles on glassy carbon electrodes. Anti-*C. jejuni* antibodies were subsequently adhered to the nanoparticles through adsorption. The incorporation of the nanoparticles significantly increased the accessible electroactive surface area for detection. *C. jejuni* from patient stool samples was analyzed using this method, and the bacterial levels were found to be in good agreement with the PCR measurements.

The ease of antibody assembly through nonspecific adhesion is an attractive strategy for the formation of electrochemical immunosensors, especially those using impedimetric detection.

9.2. Biotin–Streptavidin Antibody Immobilization

The strong binding of biotin to streptavidin or neutravidin, a deglycosylated version of avidin, enables the controlled immobilization of antibodies on electrode surfaces. The binding constant (K_D) of the biotin–streptavidin interaction is on the order of 10^{-14} M, making it one of the strongest noncovalent interactions in nature.[195] This method of antibody immobilization has proved exceptionally useful for the development of impedimetric immunosensors for a variety of targets,[196,197] including bacteria.

9.2.1. Gold Electrodes. *E. coli* have been sensitively detected on macroscopic gold electrodes through the immobilization of anti-*E. coli* polyclonal antibodies. Jaffrezic-Renault and co-workers initially formed a biotin–neutravidin pair using antibodies modified with a biotin thiol moiety and an electrode coated in immobilized neutravidin (Figure 10).[198] Impedance measurements were made to detect both whole and lysed cells using polarization resistance (R_P) as a proxy for cell binding. LODs of 10 CFU/mL for whole cells and 10^3 CFU/mL for lysed cells were found, which were compared to the LOD of 10^7 CFU/mL that was found using surface plasmon resonance (SPR) detection on the same modified gold surfaces. This demonstrates that impedance measurements to detect *E. coli* were significantly more sensitive than SPR measurements on the same surface.

Samiter and co-workers directly compared the impedimetric detection of *E. coli* O157:H7 using multiple antibody-

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immobilization strategies on a gold-disk electrode.[199] Initially, antibodies were chemically immobilized by activating a monolayer containing terminal carboxylic acids with carbodiimide prior to the addition of the antibodies. The activated carboxylic acid could react with any exposed amines on the antibody, meaning that many antibodies were in conformations that prevented their binding to bacteria. The second strategy involved biotin−neutravidin interactions using antibody biotinylation. Though this strategy can also attach to any exposed amine, the authors reported that it forms a more reproducible antibody monolayer than simple chemical adhesion. In both cases, the monolayers formed were of sufficiently high quality to block ferrocyanide. Using these antibody layers and extrapolating the R_{CT}, as low as 10−100 CFU/mL of *E. coli* were detected. Both sensor fabrication strategies demonstrated high sensitivity to *E. coli*.

Impedimetric immunosensors based on antibody immobilization with biotin−neutravidin have been extended to bacterial detection from human saliva.[192] *Streptococcus pyogenes* is a human pathogen that leads to both invasive and noninvasive infections. Disposable gold electrodes were modified with polytyramine coatings that were subsequently modified with biotin to allow for neutravidin immobilization. Biotin-tagged anti-*S. pyogenes* antibodies were then immobilized. In cumulative incubations, the linear range of detection was 100−10^5 CFU/mL, and for single incubations, the dynamic range was 100−10^4 CFU/mL. These bacteria were also selectively detected from human saliva samples.

9.2.2. Graphene Electrodes. Improving electrode architectures, namely the incorporation of three-dimensional components, has also enabled the sensitive detection of *E. coli* through antibody immobilization using biotin−streptavidin interactions. Ying and co-workers reported the development of a low-cost sensor fabricated from AuNPs immobilized on graphene paper.[200] Streptavidin assembled on the AuNPs enabled the immobilization of biotinylated anti-*E. coli* O157:H7 antibodies. Using this substrate, *E. coli* O157:H7 was sensitively detected by EIS with an LOD of 1.5×10^2 CFU/mL and a dynamic range of 1.5×10^2−1.5×10^7 CFU/mL. Because of the flexibility of the graphene paper substrate, this platform was also found to be especially tolerant to mechanical stress.

9.3. Self-Assembled Monolayers

Self-assembled monolayers (SAMs) offer unique access to chemical surface modifications. This self-assembly of monolayers can occur on a variety of substrates but is most commonly associated with silanes assembling on ITO or glass and thiols self-assembling on gold surfaces.

9.3.1. Indium Tin Oxide Coated Glass. ITO-coated glass is readily modified by chemical silanization, which can enable the further attachment of reactive handles onto which biomolecules can be coupled. For example, the Li group developed a biosensor for the specific detection of *E. coli* using antibody-modified ITO electrodes.[201] To do this, they initially modified ITO with (3-glycidoxypropyl)trimethoxysilane (GPTMS) to install the terminal epoxy groups that were subsequently reacted with anti-*E. coli* O157:H7 antibodies to generate antibody-coated electrodes. Bacterial detection was performed by measuring the electron transfer resistance of an $Fe(CN)_6^{3-/4-}$ redox probe. The LOD for the *E. coli* O157:H7 target was found to be 6000 CFU/mL, with a linear detection range from 6×10^4 to 6×10^7 CFU/mL. The Li group further characterized the antibody-modified ITO electrodes by atomic force microscopic (AFM) surface measurements and macroscopic impedance measurements.[202] The antibody monolayer was found to be homogeneous, with *E. coli* detection based on changes to both the R_{CT} and the Warburg impedance, another component of the faradic circuit model.

9.3.2. Nanoporous Membranes. In another example using SAMs, anodized alumina membranes with uniform 100 nm pores were chemically modified for the detection of *E. coli* O157:H7.[200] Initial silanization with GPTMS served as a handle for the subsequent attachment of hyaluronic acid (HA), which could be converted to a series of sulfo-NHS esters. The NHS esters reacted rapidly with anti-*E. coli* antibodies for the specific detection of this bacterium. Normal impedance change (NIC) from the ionic impedance through the pores was applied for cell quantification. The regression equation at 1 kHz for the NIC enabled bacterial concentration measurements. An LOD of 10 CFU/mL was obtained, with a dynamic range from 10 to 10^5 CFU/mL. The specificity of the sensor was confirmed using *S. aureus*, *Bacillus cereus*, and *E. coli* DH5α. Furthermore, with this sensor, the pathogenic *E. coli* O157:H7 could be detected from milk samples down to a concentration of 85 CFU/mL, making this platform a potential field sensor for foodborne pathogens.

9.3.3. Carbon-Based Electrodes. Carbon-based electrodes offer unique advantages over many other electrode materials, including a low cost and a large amount of flexibility in which supports can be used.

Importantly, carbon can be used to formulate conductive paint, enabling the construction of especially low-cost devices. In one recent example, Di Lorenzo and co-workers report an impedimetric sensor with the lectin concanavalin A (Con A) as the recognition element for bacteria.[203] The sensor was constructed by screen printing conductive ink onto hydrophobic paper. The carbon in the ink was then electrochemically oxidized to generate exposed carboxylic acid groups that could be activated with NHS and EDC to facilitate coupling of the Con A. Bacteria were successfully detected from water samples, with a limit of detection of 1.9×10^3 cells/mL and a linear range of 10^3−10^6 cells/mL. This device is portable, inexpensive, and highly sensitive, making it especially useful for bacterial detection in field applications.

In another recent example, Boukherroub and co-workers reported the construction of a sensor for the uropathogenic *E. coli* strain UTI89 that was constructed from gold electrodes modified with thin films of reduced graphene oxide/polyethylenimine.[204] The reduced graphene oxide provides an especially high surface area, and the polyethylenimine affords a large number of surface-exposed amines. The reduced graphene oxide/polyethylenimine was then modified with antifimbrial *E. coli* antibodies through amide bond formation. This sensor displayed exceptional sensitivity, with an LOD of 10 CFU/mL and a linear range of detection of 10−10^4 CFU/mL. Importantly, it functioned in water, serum, and urine and could discriminate between wild type *E. coli* UTI89 and UTI89 Δ*fim* lacking the *fim* operon, demonstrating its potential as a point-of-care diagnostic.

9.3.4. Gold Electrodes. *9.3.4.1. NHS Ester.* The chemical immobilization of anti-*E. coli* O157:H7 antibodies on gold electrodes was performed by forming a SAM of mercaptoacetic acid (MACA) wherein the terminal carboxylate was activated to an NHS ester to facilitate direct antibody coupling.[205] Following antibody layer formation, *E. coli* were detected by

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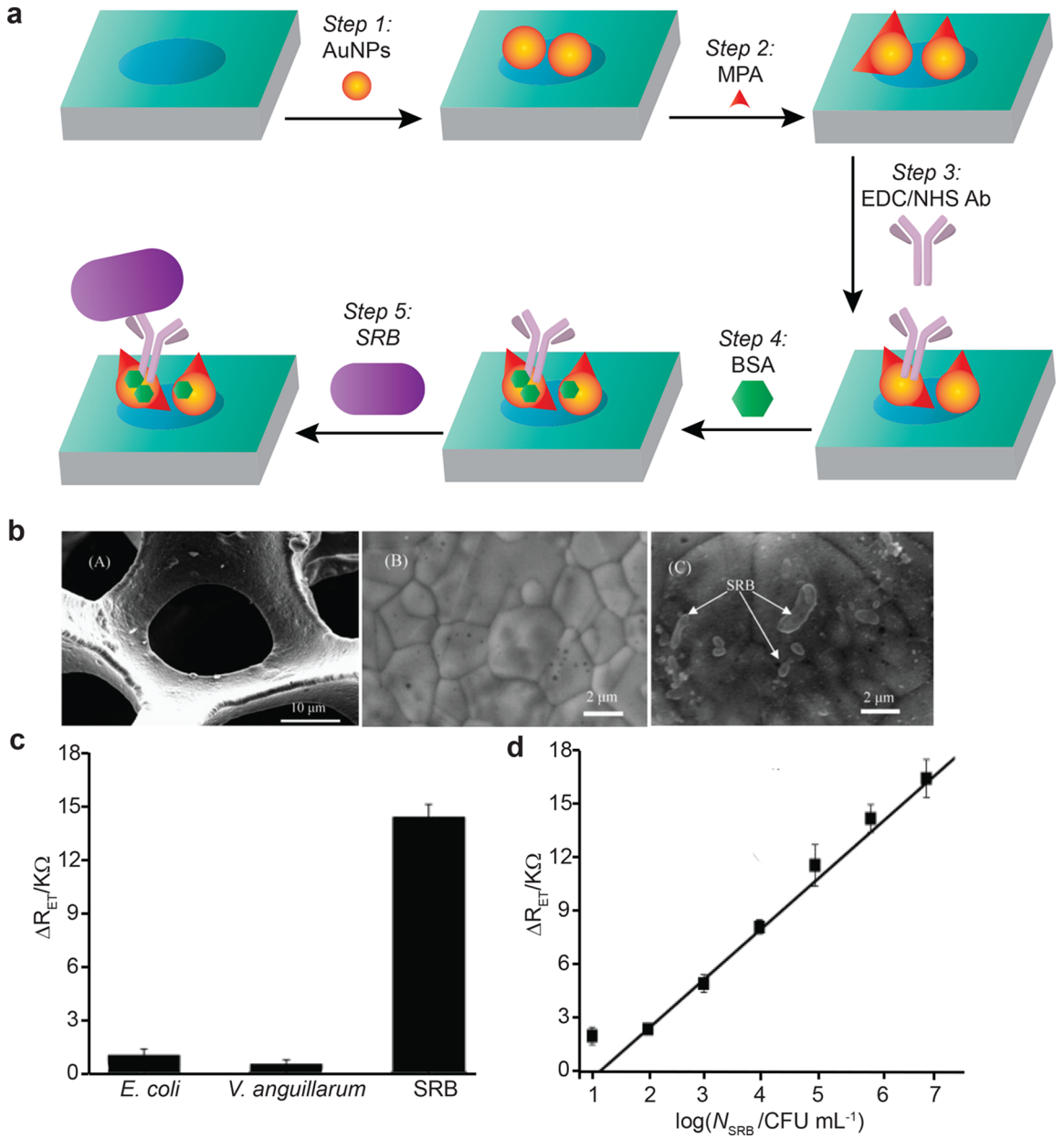


Figure 11. Three-dimensional Ni foam based impedimetric immunosensor. (a) Three-dimensional antibody-modified Ni foam electrodes are constructed in a stepwise fashion to detect sulfate-reducing bacteria (SRB). (b) SEM images are shown for Ni foam (A), antibody-modified foam without bacteria added (B), and with bacteria added (C). (c) The sensor achieves selectivity for SRB over *E. coli* and *V. anguillarum* based on the differences in the change in R_{ET}. (d) A linear range of detection for SRB was observed. Adapted with permission from ref 211. Copyright 2010 Elsevier.

EIS in the presence of $[Fe(CN)_6^{3-}]/[Fe(CN)_6^{4-}]$. The LOD was 1×10^3 CFU/mL, and the dynamic range of this sensor was 3×10^3–3×10^7 CFU/mL.

Salmonella typhimurium was similarly detected by Joshi and co-workers with an impedimetric immunosensor fabricated on NHS ester terminated monolayers.[206] They first fabricated gold electrode systems on disposable printed circuit boards, then a SAM of 16-mercaptohexadecanoic acid (16-MHDA) was formed on the gold surfaces followed by carboxylic acid activation by NHS. A monoclonal antibody against the liposaccharide surface of this species of *Salmonella* was then immobilized on the electrode. The LOD was found to be 100 CFU/100 μL, and the speed of detection with this platform was especially impressive, with successful detections occurring in under 90 s.

9.3.4.2. 3,3′-Dithiobis(sulfosuccinimidylpropionate) (DTSSP). To address the ongoing debate about the optimal method for assembling antibodies on electrodes, Pingarrón and co-workers directly compared the chemical attachment of antibodies with the homobifunctional cross-linker 3,3′-dithiobis(sulfosuccinimidyl propionate) (DTSSP) to the self-assembly of thiolated antibodies on gold screen-printed electrodes (SPEs).[207] *E. coli* detection was accomplished through EIS monitoring of R_{CT} changes in the presence of $[Fe(CN)_6^{3-}]/[Fe(CN)_6^{4-}]$. The LOD of this sensor was just over 3 CFU/mL, with a linear range of 5–10^8 CFU/mL. Importantly, the sensor was specific for *E. coli* over *S. aureus*

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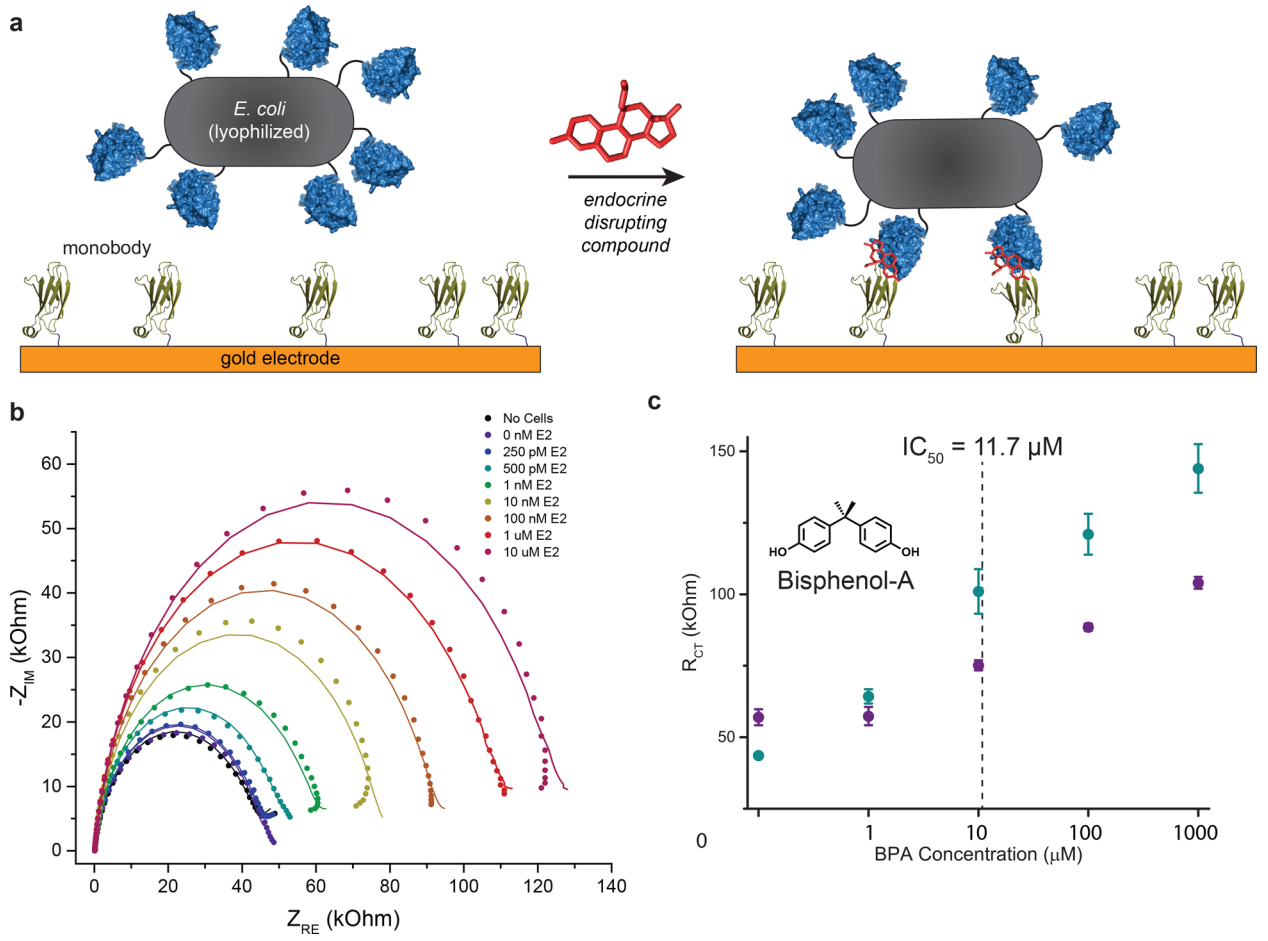


Figure 12. Bacterial impedimetric sensor for pollutants. (a) A sandwich assay based on bacterial binding to an electrode is used to detect pollutants. *E. coli* modified with a mammalian estrogen receptor only binds to gold electrodes modified with a monobody in the presence of a small molecule pollutant. (b) Experimental impedance data and Warburg circuit model fits are shown for the detection of the native hormone, estradiol. (c) The detection of bisphenol A (BPA) in buffer (blue) and infant formula (purple) was achieved with the sensor at sensitivities below the IC_{50} of BPA. Adapted from ref 45. Copyright 2017 American Chemical Society.

and *Salmonella choleraesuis*, and *E. coli* was detected from inoculated tap and river waters with an LOD of 10 CFU/mL.

9.3.4.3. Hyaluronic Acid. In addition to simple monolayer formation on gold, adding complexity to the SAM has been shown to reduce nonspecific binding and improve capture efficiency for impedimetric immunosensors. A gold electrode was modified with a hydrophilic layer of hyaluronic acid prior to antibody immobilization.[208] *E. coli* O157:H7 was selectively detected by changes in the NIC of the electrode at 0.1 Hz in the presence of $[Fe(CN)_6]^{3-/4-}$. The LOD of this sensor was 7 CFU/mL with a dynamic range of $10-10^5$ CFU/mL.

9.3.4.4. Conductive Polyaniline. The versatility of gold electrodes has also enabled the immobilization of conductive polymers for antibody conjugation. Sen and co-workers developed an impedance immunosensor for *E. coli* O157:H7 through the initial formation of a polyaniline (PANI) layer on a gold electrode followed by glutaraldehyde coupling to conjugate the anti-*E. coli* antibody.[209] With impedance-based detection, an LOD of 10^2 CFU/mL was obtained with a dynamic range up to 10^7 CFU/mL.

For all available methods of surface modification, it is essential to run comparative studies to establish the best technique for a given application. Recently, Albanese and co-workers reported an impedimetric immunosensor for the specific detection of *E. coli* O157:H7.[210] Multiple strategies for anti-*E. coli* antibody immobilization on gold electrodes were compared for sensitivity and selectivity, including a non-oriented antibody layer on a monolayer of 4-mercaptobenzoic acid (MBA) with NHS chemistry, oriented layers of antibody assembled on MBA monolayers incubated with Protein A/G, coupling to electrodeposited cysteamine layers, ferrocene carboxylic acid sandwiched between cysteamine layers followed by antibody immobilization, and finally, a similar strategy to ferrocene linkages but with an additional layer of polyamidoamine (PAMAM) dendrimers. Each of these surface modification strategies yielded a different detection limit and linear range of detection. The lowest detection limit, 3 CFU/mL, was achieved on cysteamine/ferrocene-modified electrodes. The authors cite the importance of incorporating an electron transfer intermediate, ferrocene, for the low limits of detection and relatively large linear detection ranges for the two most successful strategies (cysteamine/ferrocene and cysteamine/PAMAM/ferrocene). Furthermore, *E. coli* doped into meat and milk samples were successfully quantified with several of the sensors, and the results were found to be in good agreement with ELISA analyses of the same food samples. This study, with direct comparisons between immobilization strategies, demonstrates how small changes can affect detection limits.

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9.3.5. Nickel Foam. Sulfate-reducing bacteria (SRB) are an important class of bacteria that can rapidly contaminate environments. Hou and co-workers developed an impedance-based immunosensor for SRB on antibody-modified Ni foam (Figure 11).[211] The 3D Ni foam was modified in a stepwise manner to immobilize antibodies on the platform; an initial AuNP modification was followed by the assembly of 11-mercaptoundecanoic acid (MPA) and activation of the terminal carboxylates with NHS. Anti-SRB antibodies were immobilized on this platform, and the remainder of the surface was passivated with BSA. Monitoring SRB binding by R_{CT} changes gave a large dynamic range of 2.1×10^1–2.1×10^7 CFU/mL. This 3D platform offers an alternative scaffold to conventional planar electrodes for bacterial detection.

9.4. Impedance Immunosensor for Pollutant Detection

In a variation of a conventional impedance immunosensor for bacterial detection, we have developed an impedance sandwich assay for the detection of small molecule endocrine pollutants that uses a bacterial cell as part of the sandwich assay (Figure 12).[45] One half of the sandwich consists of the native human estrogen hormone receptor ERα expressed on the surface of *E. coli* through fusion with an ice nucleation protein. The second half of the sandwich is a monobody self-assembled on a disposable gold electrode that only binds ERα in the presence of a small molecule pollutant. Large responses in the electrode R_{CT} were observed in the presence of sub-parts-per-billion endocrine disruptors. As the ERα is scaffolded on the surface of the *E. coli*, the size of the bacteria provides inherent signal amplification. Estrogenic compounds were detected from complex sample matrixes, including infant formula.

10. APTAMER-BASED DETECTION

As discussed previously, aptamers offer some advantages over proteins and antibodies. Their ability to tolerate a broader range of solvents than proteins and the rapid, low-cost synthesis of chemically modified DNA sequences make them appealing for applications in impedance-based biosensors, especially for point-of-care technologies.[197] Though many impedimetric aptasensors have been developed, the majority are sensors for either particular DNA sequences or proteins,[212] though in some cases, these are proteins or DNA sequences specific to pathogenic bacteria.[213,214] The detection of whole bacteria using these methods has been fairly limited, though examples do exist.

10.1. Screen-Printed Carbon Electrodes Modified with AuNPs

10.1.1. AptaVISens-B. The AptaVISens-B platform developed by the Berezovski group is an aptamer-based viability impedimetric sensor for bacteria.[215] It is based on their AptaVISens-V platform for the aptamer-based sensing of viruses.[216] Initially, an aptamer was selected for live *S. typhimurium* by Cell-SELEX.[50] Positive selections were performed against viable *S. typhimurium*, while extensive negative selections were performed against heat-killed *S. typhimurium*, as well as many other pathogens, including *Salmonella enteritidis*, *Citrobacter freudii*, *E. coli*, *S. aureus*, and *P. aeruginosa*. Following isolation of a sequence specific to viable *S. typhimurium*, the aptamer was allowed to self-assemble on a screen-printed carbon electrode modified with AuNPs. When EIS was measured in the presence of $[Fe(CN)_6]^{3-/4-}$, *S. typhimurium* was sensitively and specifically detected with LODs down to 600 CFU/mL and a dynamic range between 1×10^3 and 1×10^5 CFU/mL.

10.1.2. Aptamer-Based Impedimetric Sensor for Typing of Bacteria (AIST-B). Along with a specific viability sensor, the Berezovski group also developed an aptamer-based platform for the rapid typing of bacteria using aptamers.[217] This sensor, termed aptamer-based impedimetric sensor for typing of bacteria (AIST-B), involved Cell-SELEX for the initial identification of a DNA aptamer specific to *S. enteritidis* that did not bind to other pathogenic bacteria or *Salmonella* species, including *S. typhimurium* and *S. choleraesuis*. The identified aptamer sequence was then immobilized on AuNP-modified screen-printed carbon electrodes, enabling an LOD of 600 CFU/mL. Their sensor provided a rapid readout, with detection occurring in 10 min.

10.2. Poly[pyrrole-*co*-3-carboxyl-pyrrole] Copolymer Aptamer Support

An efficient impedance-based sensor for *S. typhimurium* was constructed on gold disc electrodes through the initial electropolymerization of a copolymer of pyrrole-3-carboxylic acid and pyrrole.[218] Free carboxylic acids within the polymer were activated using NHS, followed by the addition of an amino-terminated aptamer that had been previously identified as an *S. typhimurium*-specific sequence.[219] Using this assembly, the especially sensitive detection of *S. typhimurium* was accomplished, with an LOD of 3 CFU/mL, an LOQ of 100 CFU/mL, and an impressive dynamic range of 10^2–10^8 CFU/mL.

10.3. Diazonium Aptamer Coupling

Inexpensive, disposable, screen-printed carbon electrodes were used as a substrate for either electrochemical or chemical diazonium couplings that were used as a handle for amino-modified DNA aptamers.[220] Aptamers specific to *S. typhimurium* were immobilized on the carboxylic acid terminated layer formed following the diazonium coupling. Higher density aptamer surface coverages were found to improve bacterial detection. A linear range of detection from 10 to 10^8 CFU/mL of bacteria was observed with this platform, with an LOQ of 10 CFU/mL and an LOD of 3 CFU/mL. This platform distinguished *S. typhimurium* from other *Salmonella* species and could detect the bacteria spiked into apple juice samples.

11. DNA HYBRIDIZATION-BASED CELL ADHESION

As described above, bacteria can be difficult to capture for quantification. We have facilitated this capture and impedimetric quantification of nonadherent cells on electroactive surfaces through cell immobilization using DNA hybridization-based cell adhesion.[61,62] Bacteria were modified with DNA through an initial exposure to sodium periodate. Hydrazide-modified DNA was then added to the cells in the presence of aniline to form a hydrazone linkage between the DNA sequence and oxidized sugars on the surface of the bacteria. We control the density of cells on the electroactive surfaces through the DNA surface density.[62] Studies of *S. oneidensis* found that, as mismatches were incorporated in the hybridization DNA sequence, the number of bacteria on the electrode remained constant as determined by the change in the R_{CT} value following EIS in the presence of $[Fe(CN)_6]^{3-/4-}$.[63] Impedance-based bacterial quantification enabled direct comparison to currents generated by these electron-transfer-proficient cells at the electrode surface. The ability of these cells to generate current prior to biofilm

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formation significantly speeds up the formation of *S. oneidensis*-coated electrodes for biofuel cells.[63]

12. BACTERIOPHAGE-BASED DETECTION

Because of the inherent specificity of bacteriophage for bacteria, several impedance-based platforms have been developed that rely on the bacteriophage-based capture of microbes.[221] In one of the earliest examples of bacteriophage-mediated detection, a screen-printed carbon electrode microarray was modified with bacteriophage by generating carboxylic acids on the electrode surface, which, following activation with EDC, were directly coupled to T4 bacteriophage protein coats.[222] *E. coli* were specifically detected with these modified electrodes, with an LOD of 10^4 CFU/mL and a maximum signal after 20 min of incubation. Longer incubation time periods induced bacterial lysis.

A similar strategy was used to develop a bacteriophage-based biosensor for *Listeria*.[223] To construct this biosensor, the cell wall binding domain (CBD) of bacteriophage-encoded peptidoglycan hydrolases (endolysin) was immobilized via NHS/EDC coupling on gold screen-printed electrodes modified with a SAM. *Listeria innocua* serovar 6b was selectively captured from pure media with an LOD of 1.1×10^4 CFU/mL, as well as from inoculated milk with an LOD of 10^5 CFU/mL.

An additional example involved *E. coli* detection and quantification with T4 bacteriophage modified electrodes and complementary electrochemical detection methods.[224] The T4 with EIS-based detection was initially used as a viability assay for *E. coli*. A gold electrode was modified with T4 bacteriophage through reaction with a cystamine-modified electrode in the presence of 1,4-phenylene diisothiocyanate. A complementary linear sweep voltammetry technique was used as a confirmation through loop-mediated isothermal amplification (LAMP) of the *E. coli Tuf* gene. The dynamic range for the impedimetric component was 10^3–10^9 CFU/mL, while the dynamic range for the LAMP component was 10^2–10^7 CFU/mL

13. CONCLUSIONS AND FUTURE TRENDS

Impedance-based bacterial monitoring has been a key technique over the past half-century because of its speed and ease-of-use advantages over conventional bacterial detection methods. Impedance microbiology was initially used to monitor overall bacterial growth and biofilm formation, which has led to the development of several commercial products. As the field matured, the development of sophisticated impedance-based biosensors enabled the specific detection of particular pathogens. Though impedance methods have enabled the sensitive detection of pathogenic bacteria, the field remains limited with few sensors moving to commercialization. In many cases, challenges remain in achieving necessary detection limits, especially in complex samples such as food or whole blood. Additionally, many platforms that require biomolecules for specific detection become cost-prohibitive for commercialization. As improvements to disposable electrode technology couple with reduced biomolecule production costs, these biosensors will continue to move toward point-of-care use. Advances in bioconjugation strategies and surface immobilization methods are likely to improve detection limits and specificities. As the importance of interactions between host cells and pathogenic microbes, as well as interactions between bacterial species in complex communities, is elucidated, impedance techniques will be vital for their continued study. Systems such as infected tissue and the human microbiome offer a new and important direction for the development of impedimetric sensors for the rapid, specific monitoring of bacterial species.

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Matthew B. Francis is the T. Z. and Irmgard Chu Distinguished Professor in Chemistry and the Chair of the University of California, Berkeley Chemistry Department. In addition, he is a Faculty Scientist at the Lawrence Berkeley National Laboratory. Over the years, Matt has received the Dreyfus Foundation New Faculty Award, an NSF Career Award, and a GlaxoSmithKline Young Investigator Award. He received his undergraduate degree in chemistry from Miami University in Oxford, OH. He completed his Ph.D. at Harvard University in the lab of Prof. Eric Jacobsen developing combinatorial strategies for the discovery and optimization of new transition metal catalysts. He then moved to UC Berkeley, where he was a postdoctoral fellow in the Miller Institute for Basic Research in Science. He worked under the guidance of Prof. Jean Fréchet, focusing on the development of DNA-based methods for the assembly of polymeric materials and the application of dendrimers for drug delivery.

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